Advancing sustainability with satellite data and computing: climate & health

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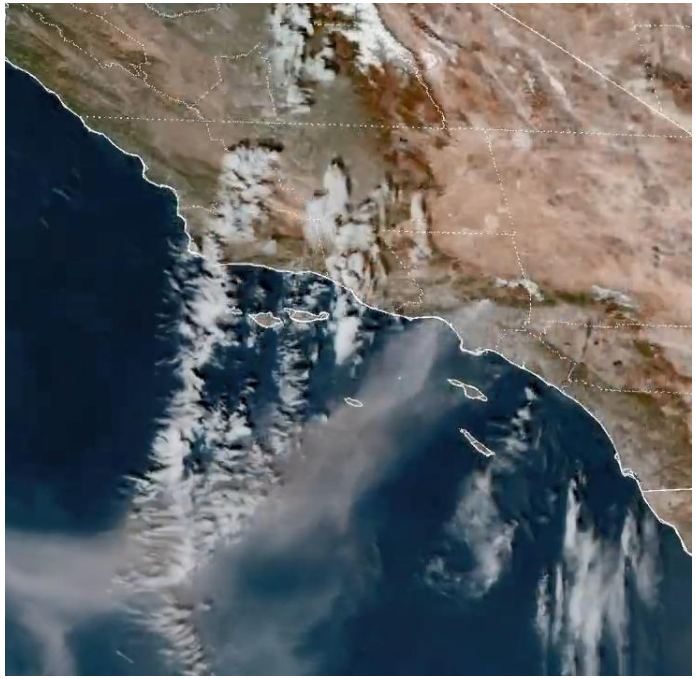
LEAP summer fellow, Columbia University

New York University

July 30, 2025

Monitor our environment with satellite data

smoke from wildfires (Los Angeles ‘25)



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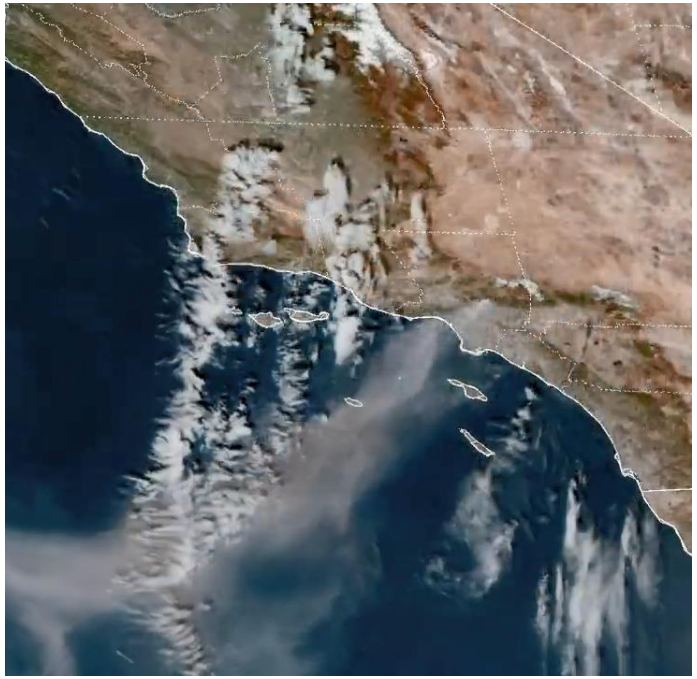
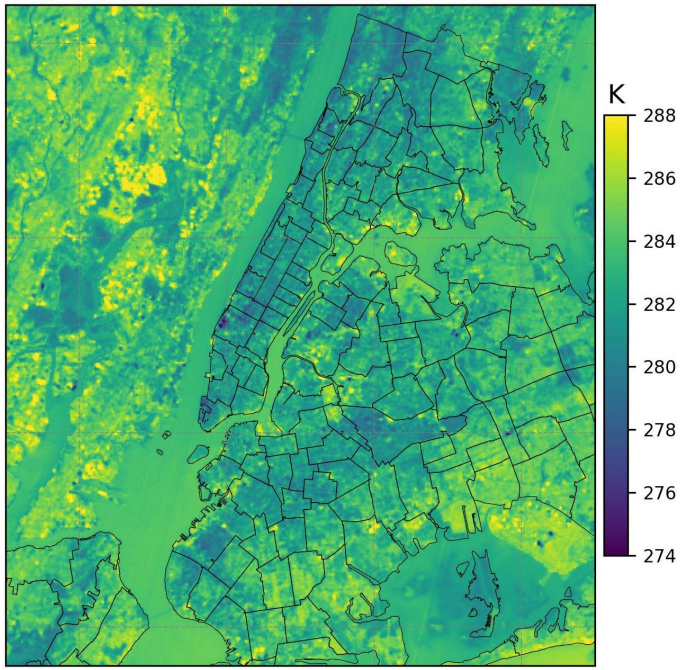


temperature (NYC)



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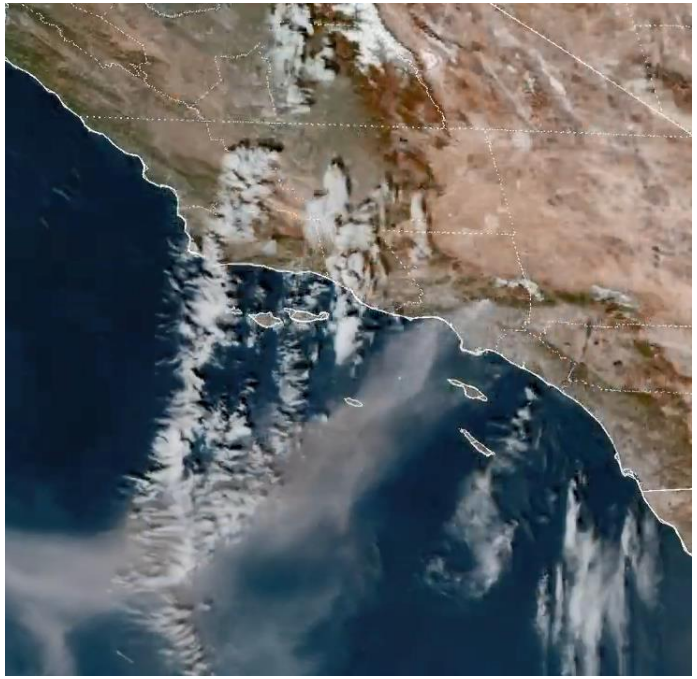

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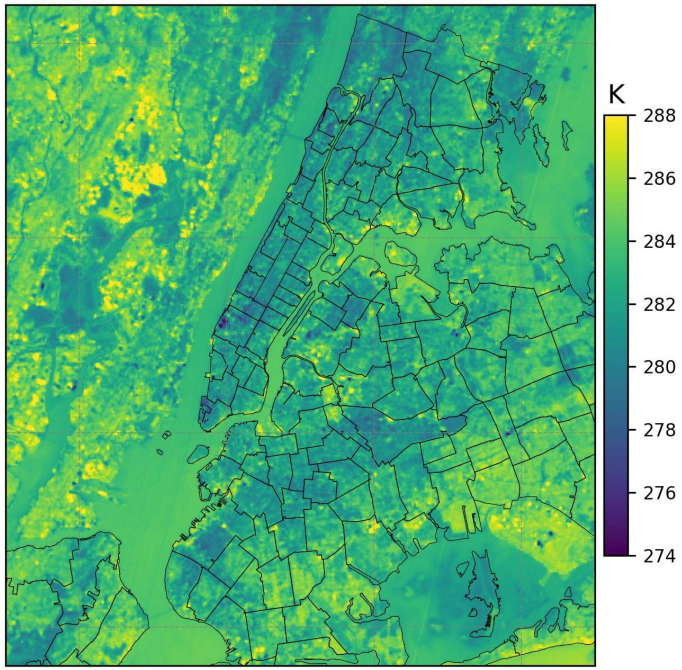
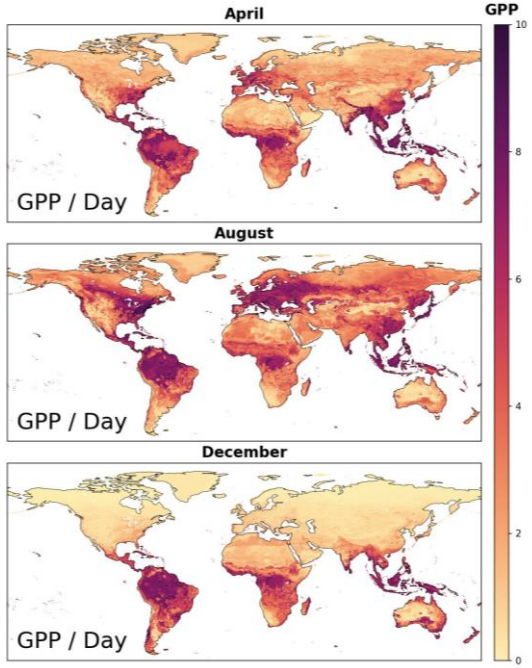
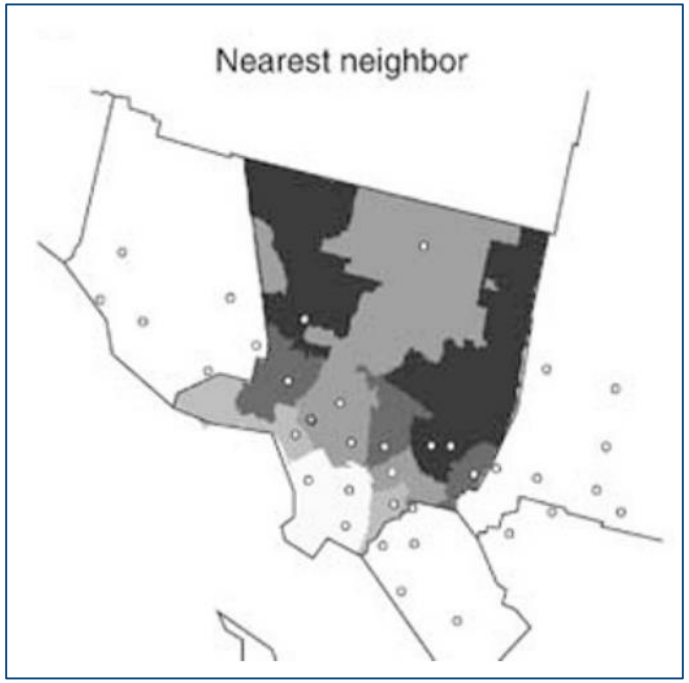


gross primary production (GPP)



Monitor without satellite data?

air quality stations (Los Angeles)



[Wong et al. 04, Gaffin et al. 08, Pastorello et al. 19]

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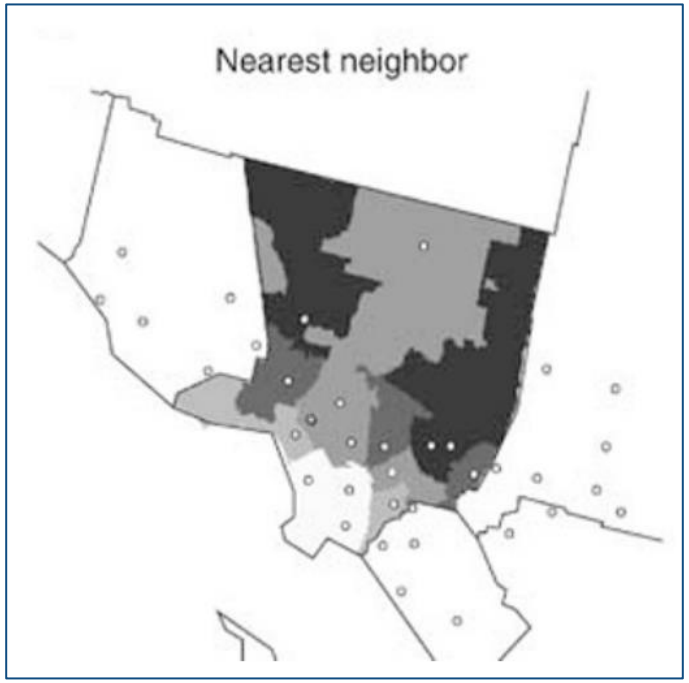
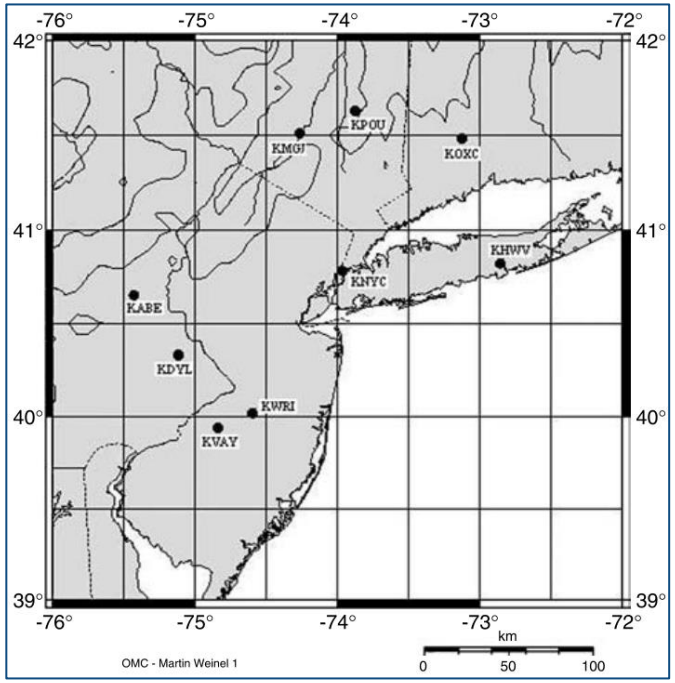


weather stations (tri-state area)



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Monitor without satellite data?

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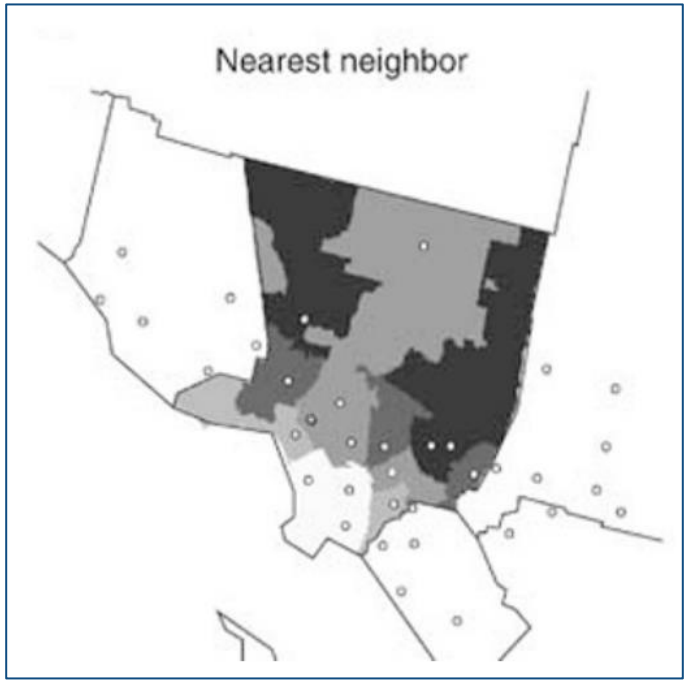

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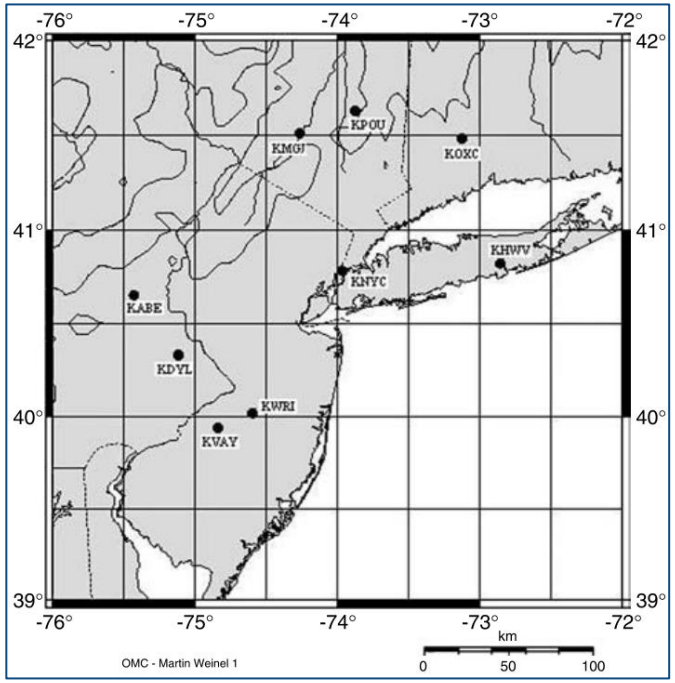
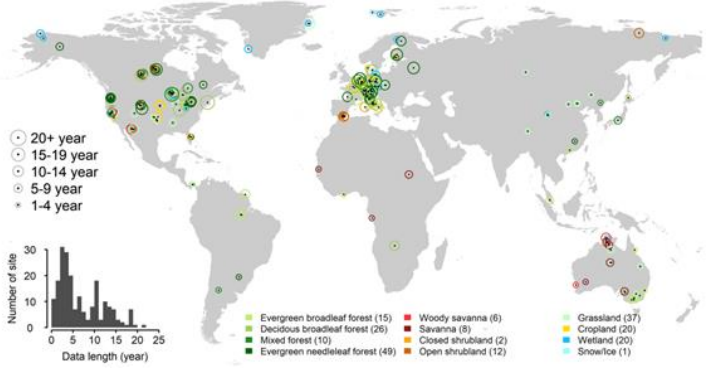


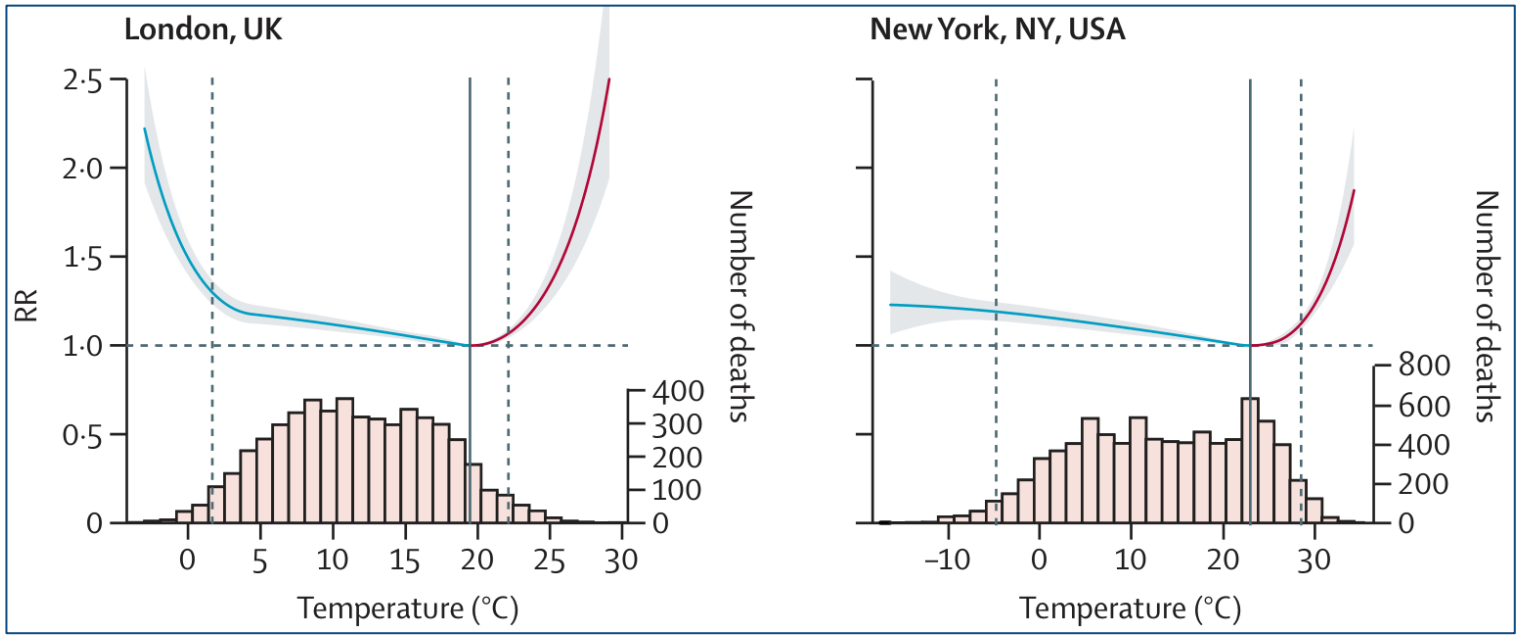
flux towers



[Wong et al. 04, Gaffin et al. 08, Pastorello et al. 19]

Climate & Health

existing climate & health research



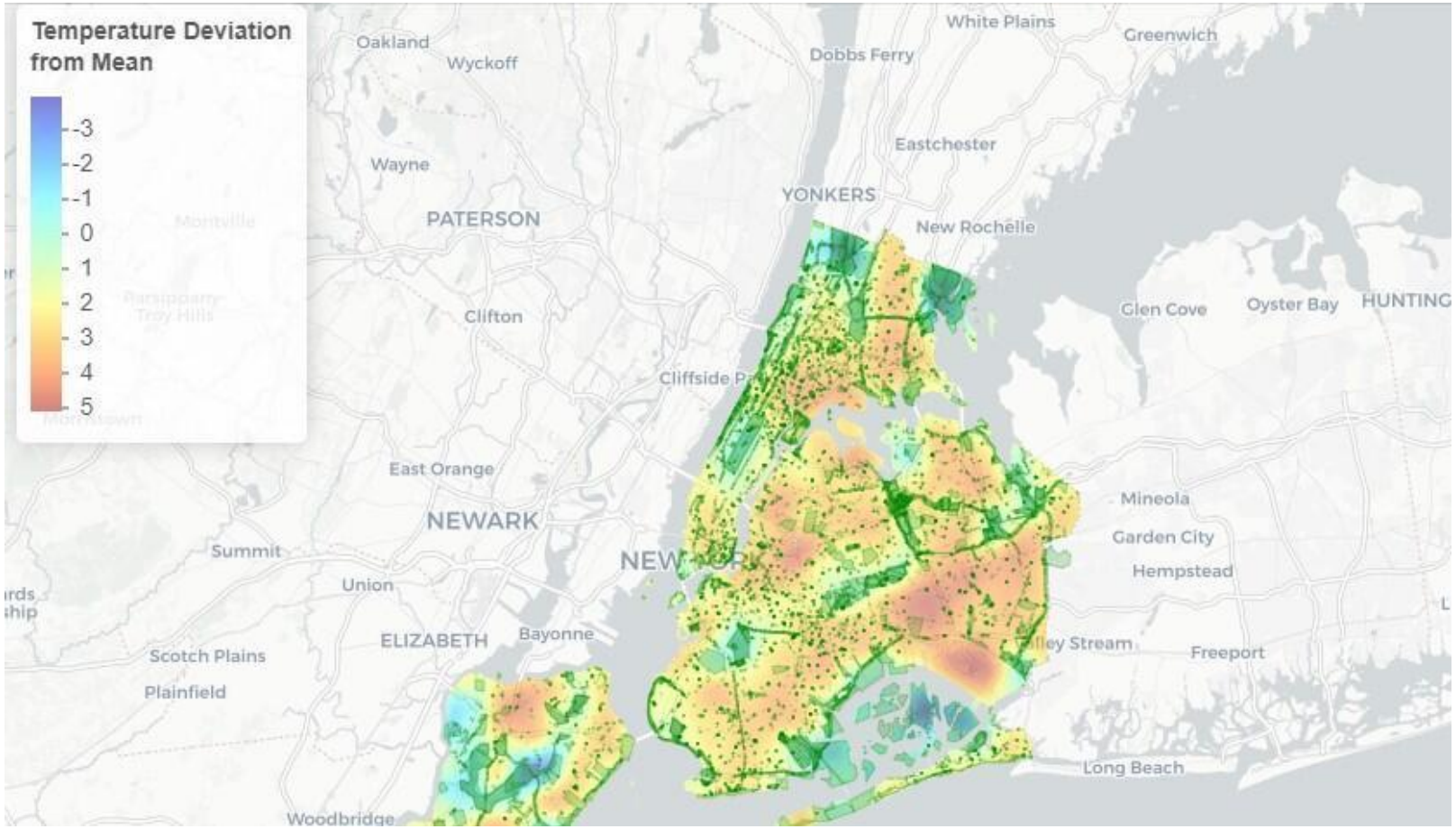
one city
one weather station
daily average

cannot determine
intraurban heterogeneity

population differences

[Gasparrini et al. 15]

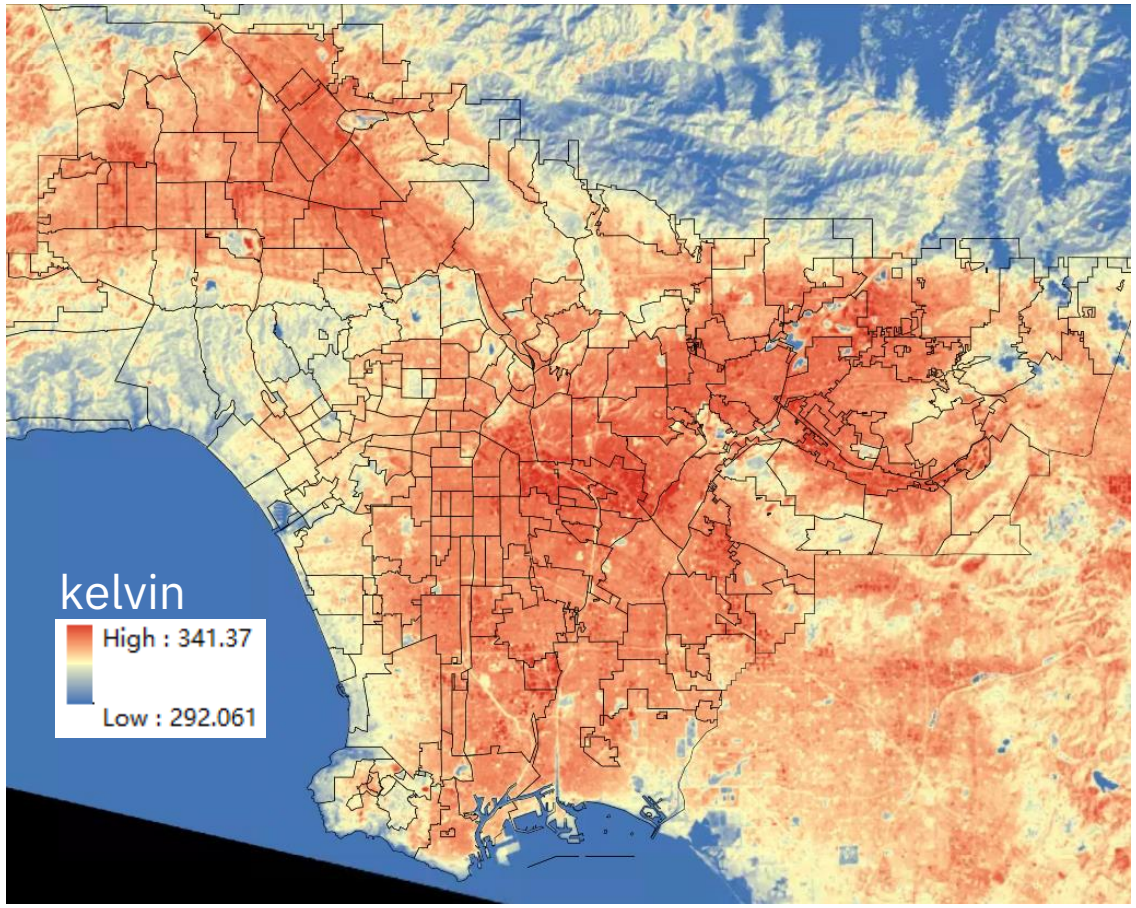
Temperature is heterogeneous



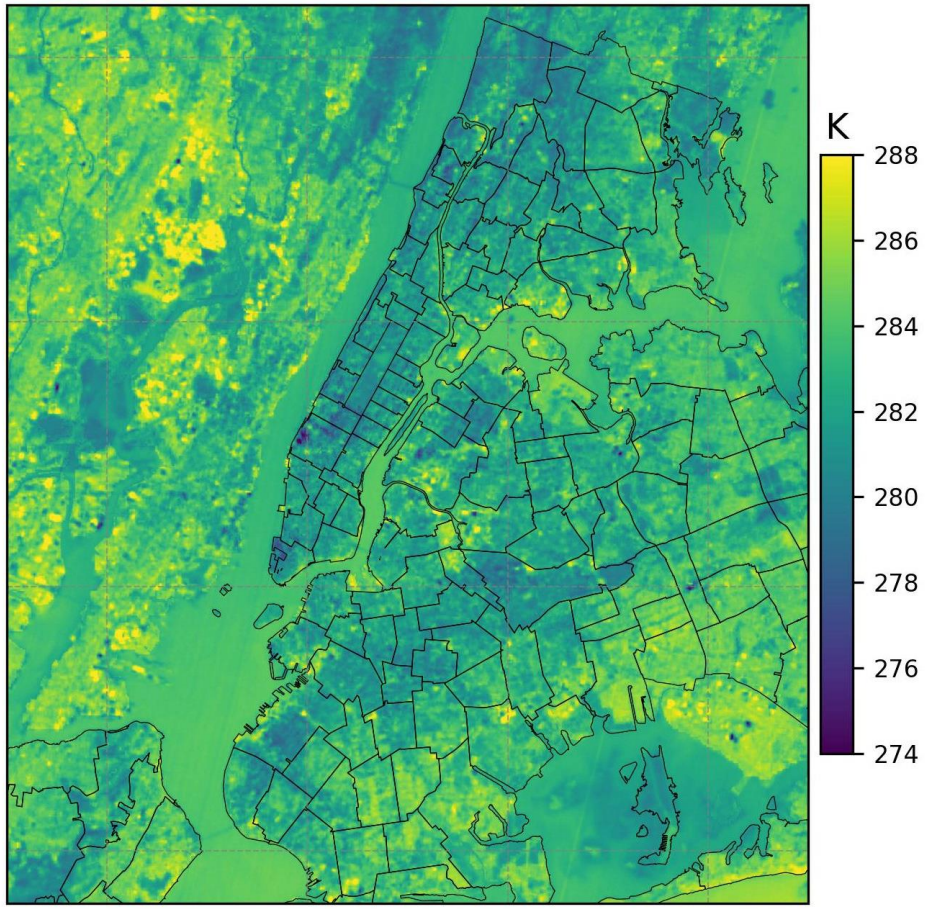
Map generated from Landsat temperature.
New York City Council's Data Team. Public Domain.

Temperature is heterogeneous

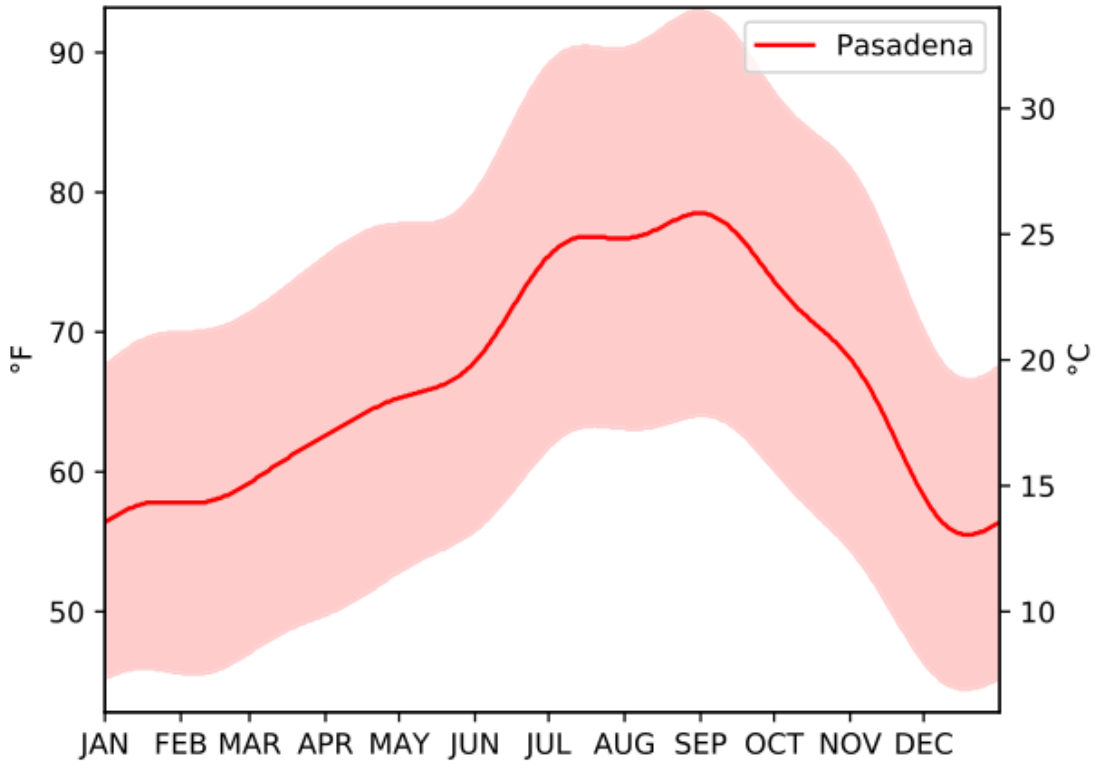
Southern California heatwave, 7 September 2024

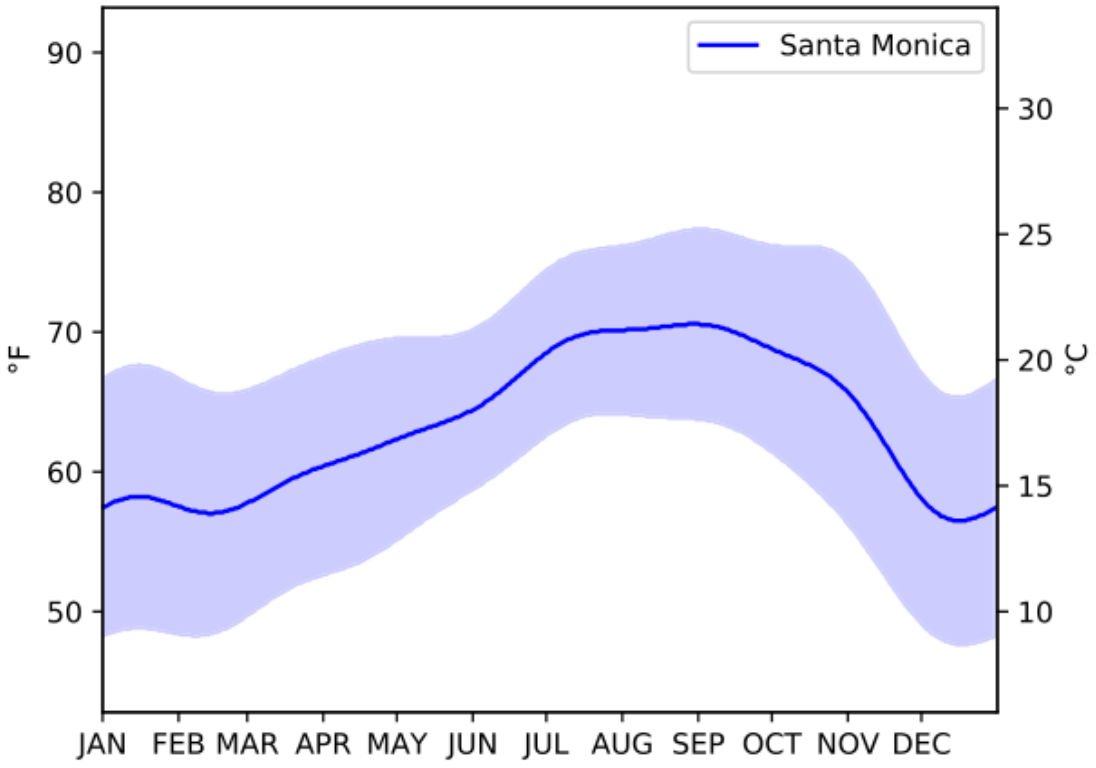


NYC on a random day



Mortality and prolonged exposure to diurnal temperature range (DTR) [Liu et al. 23]

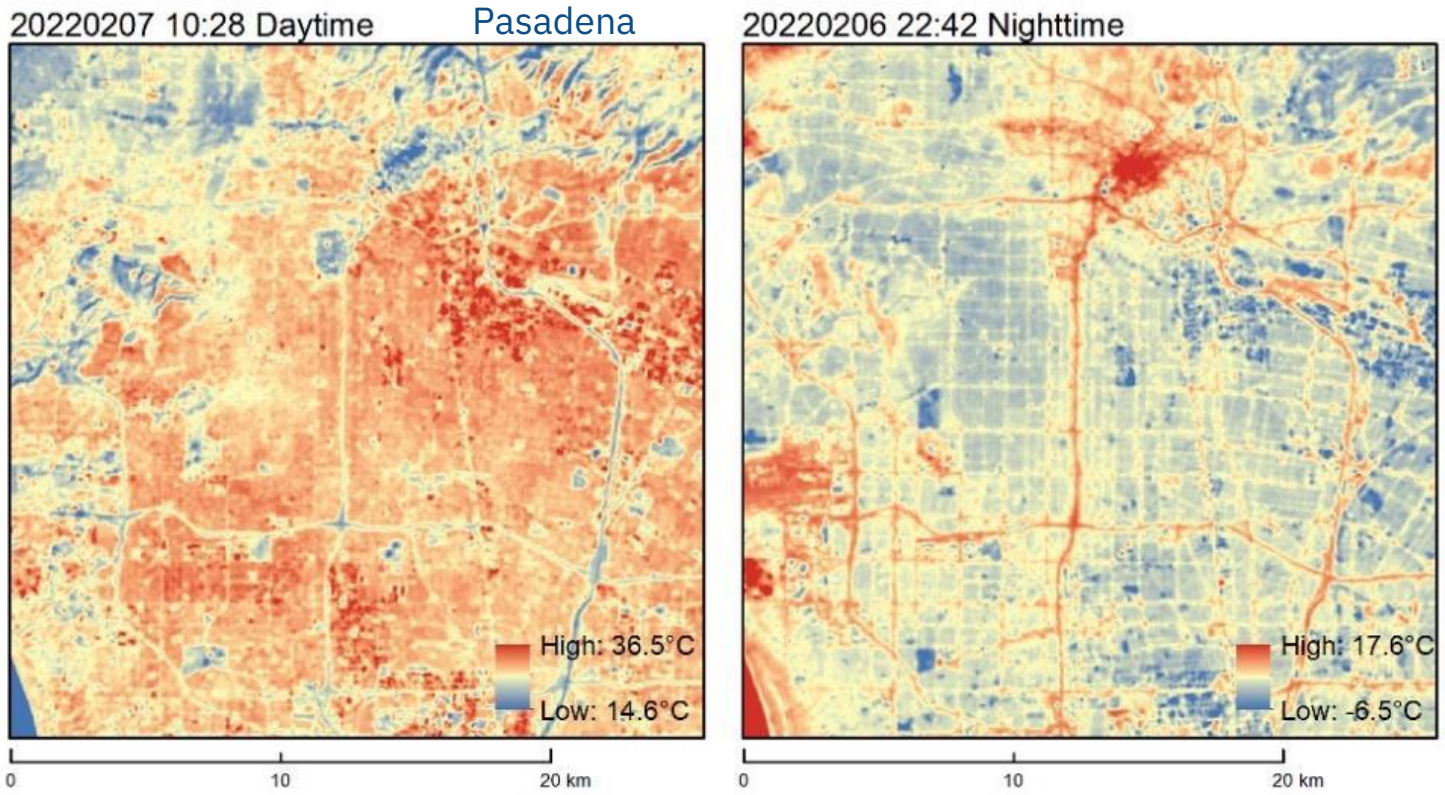




Mortality and prolonged exposure to diurnal temperature range (DTR) [Liu et al. 23]

Pasadena

Santa Monica



day night differences

(somewhat) similar pattern across the years

Large temperature swings and health in East Asian cultures

ChatGPT

In Japanese, there is a saying that conveys a similar idea about the impact of weather on health:

昼と夜の寒暖差に注意（ひるとよるのかんだんさにちゅうい）
"Hiru to yoru no kandan-sa ni chūi."

This can be translated as "Be cautious of the temperature difference between day and night." It reflects the awareness of the potential health risks associated with significant temperature fluctuations. While it may not be an exact equivalent to the Chinese saying you mentioned, the concept is similar.

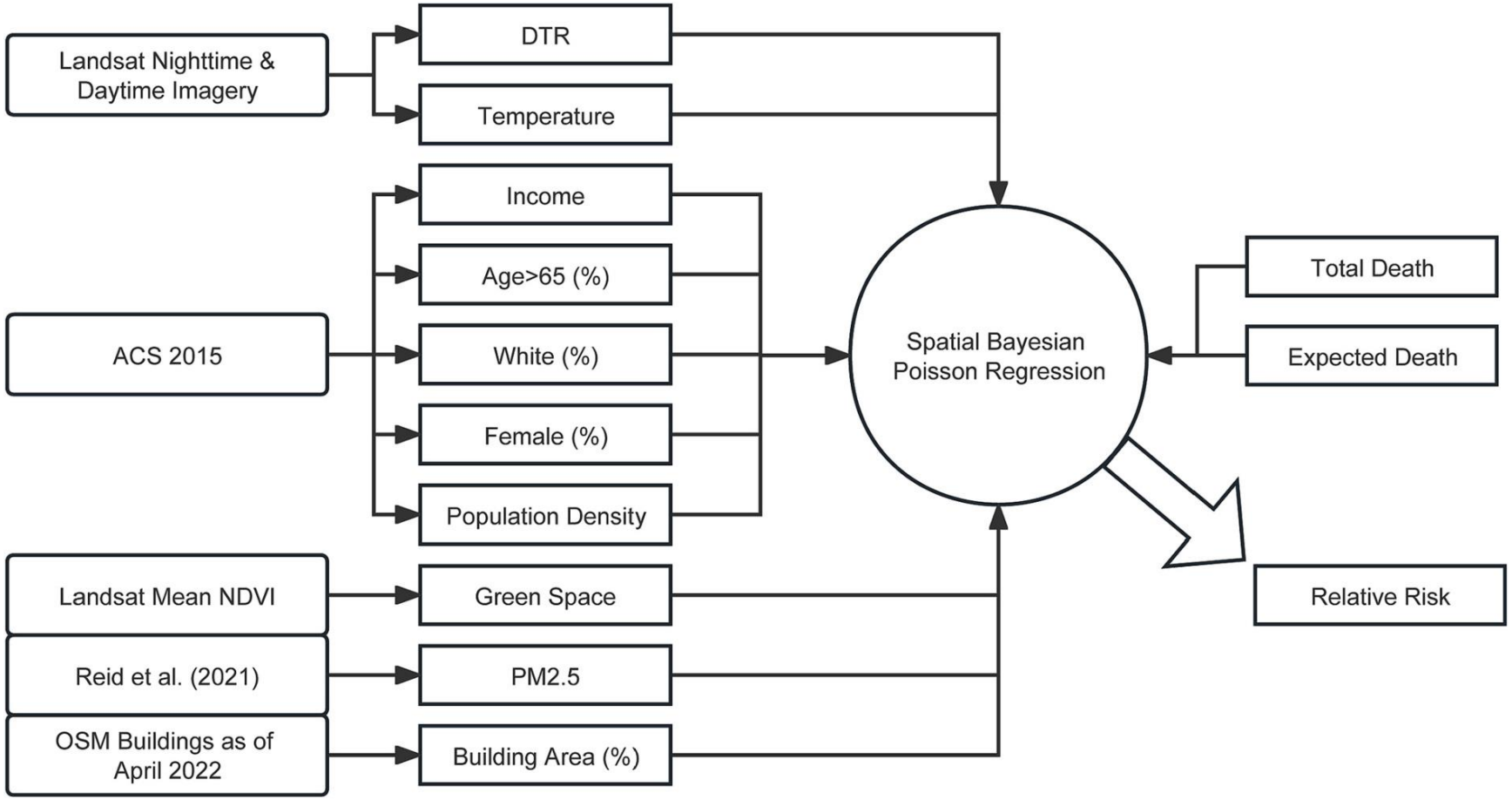
ChatGPT

In Korean, there isn't an exact equivalent saying to "昼夜温差大" (Large temperature difference between day and night). However, Koreans may express similar caution regarding health and weather changes. Here's a phrase that conveys a somewhat related idea:

"낮과 밤의 기온 차이에 주의하다" (Najgwa bamui gion chaie chuwi hada), which can be translated as "Be cautious of the temperature difference between day and night."

While this may not be a traditional proverb, it captures the essence of being mindful of temperature fluctuations, similar to the concept expressed in the Chinese saying you mentioned. Keep in mind that expressions and idioms can vary, and there might be regional or colloquial phrases that convey similar meanings.

Mortality and prolonged exposure to diurnal temperature range (DTR) [Liu et al. 23]



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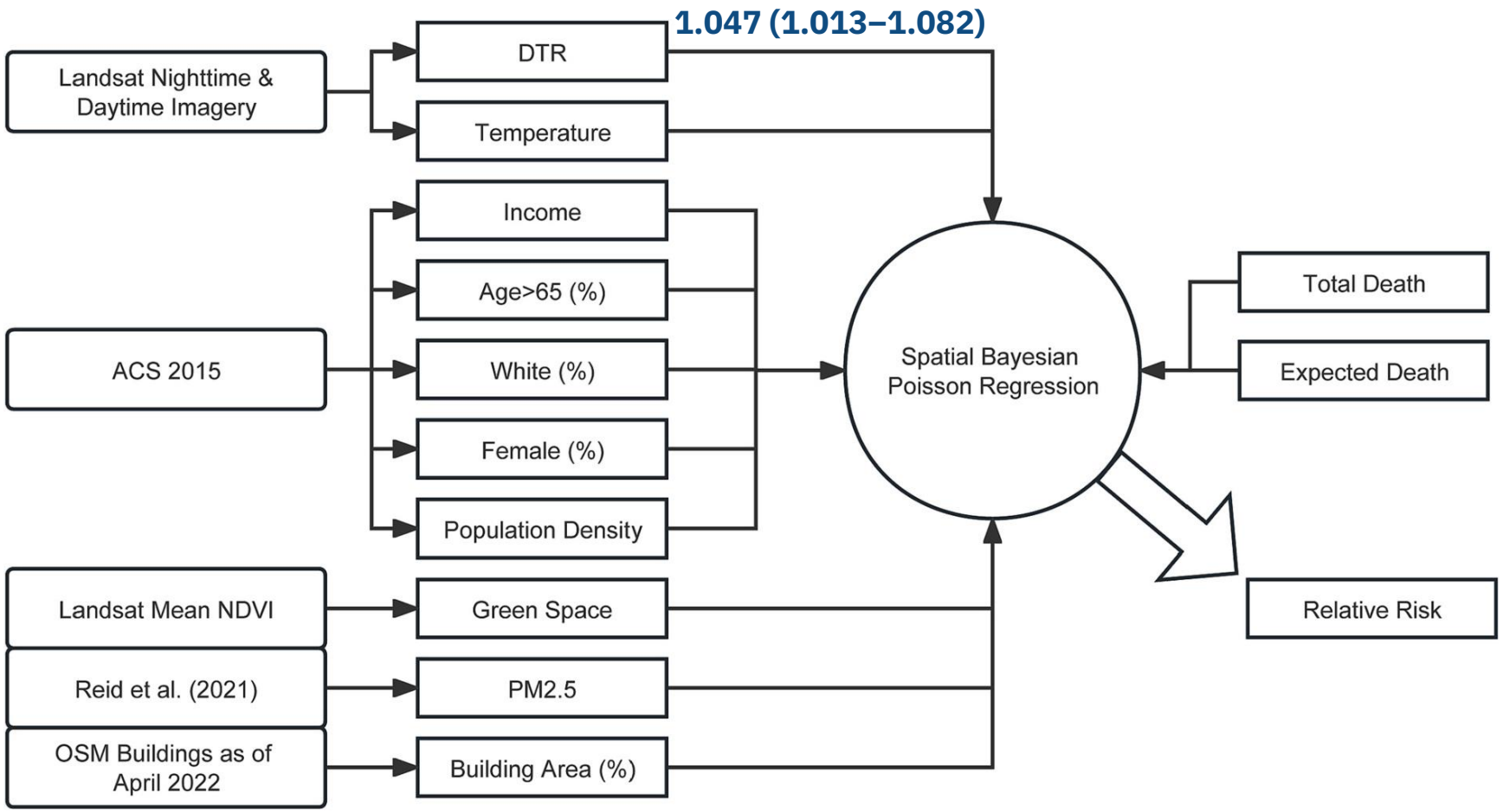

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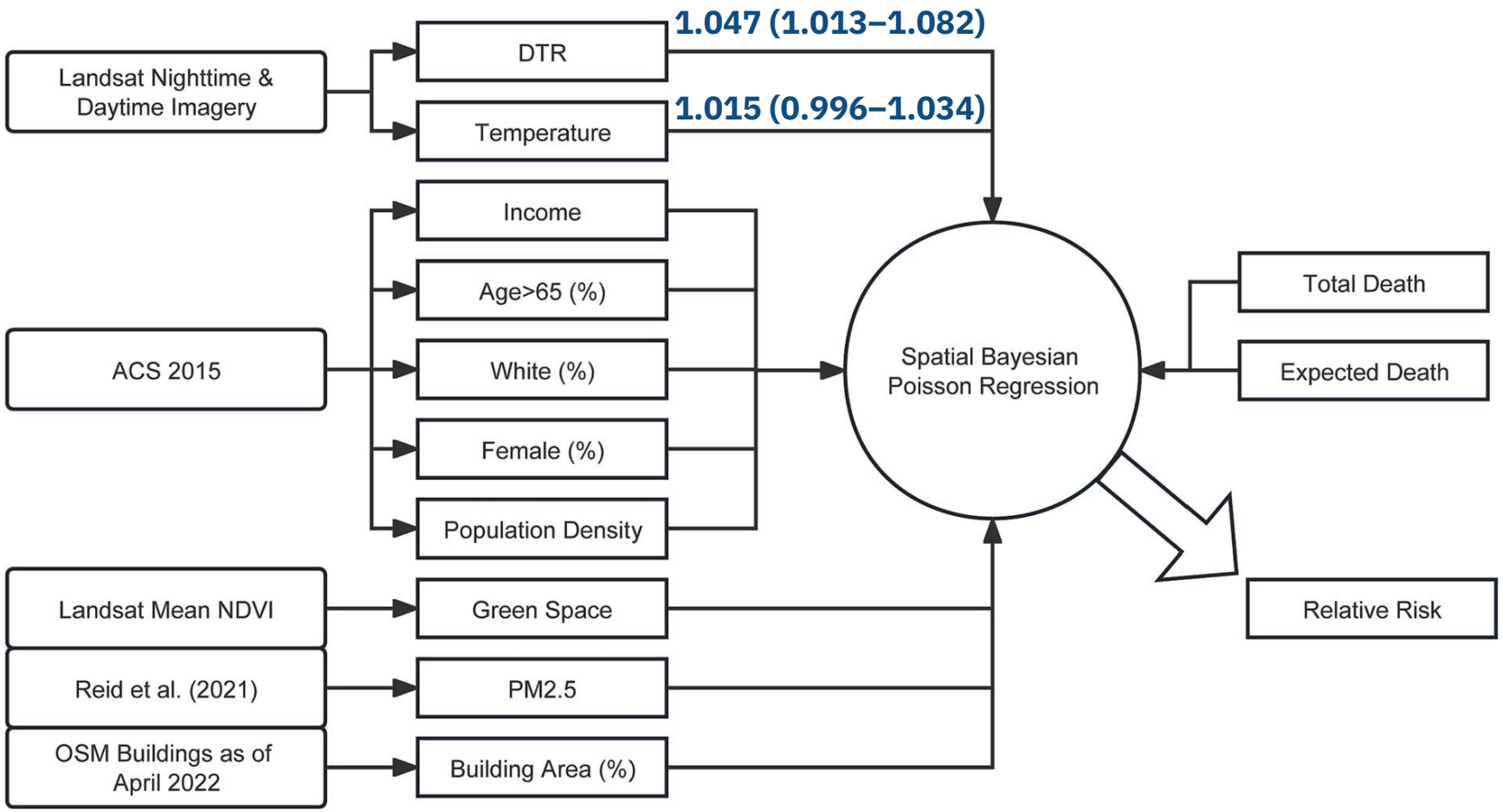

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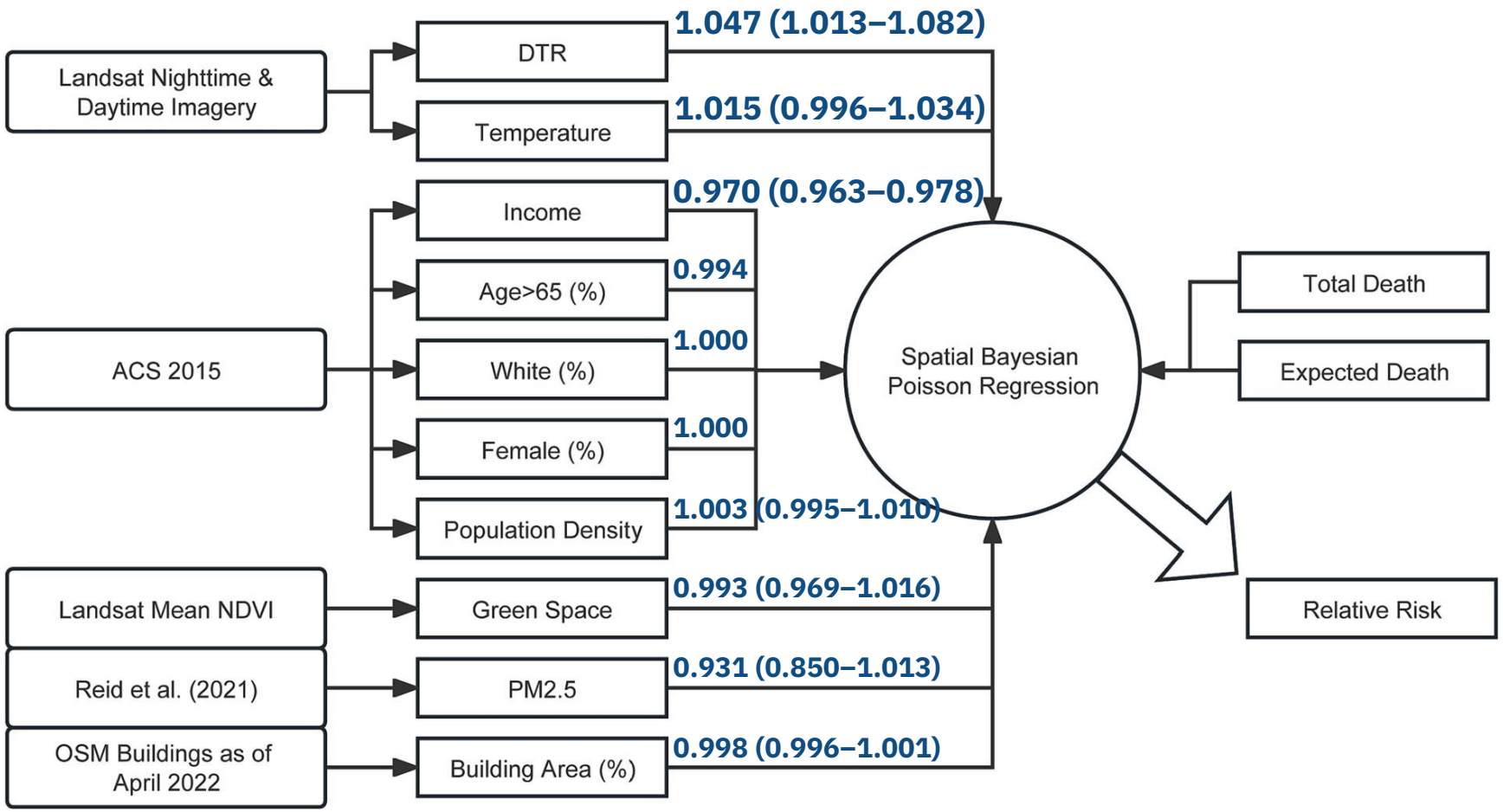
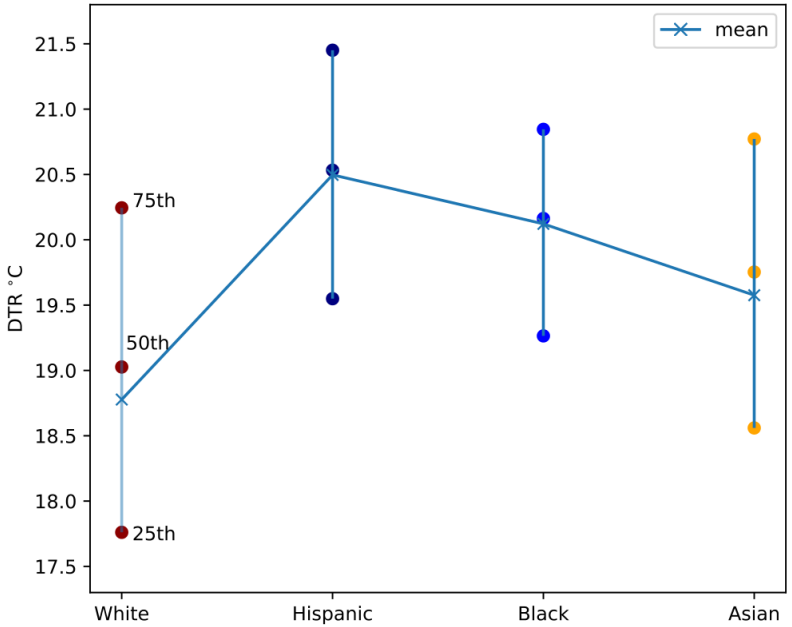


Unequal exposure by race/ethnicity and income

[Liu et al. 23]



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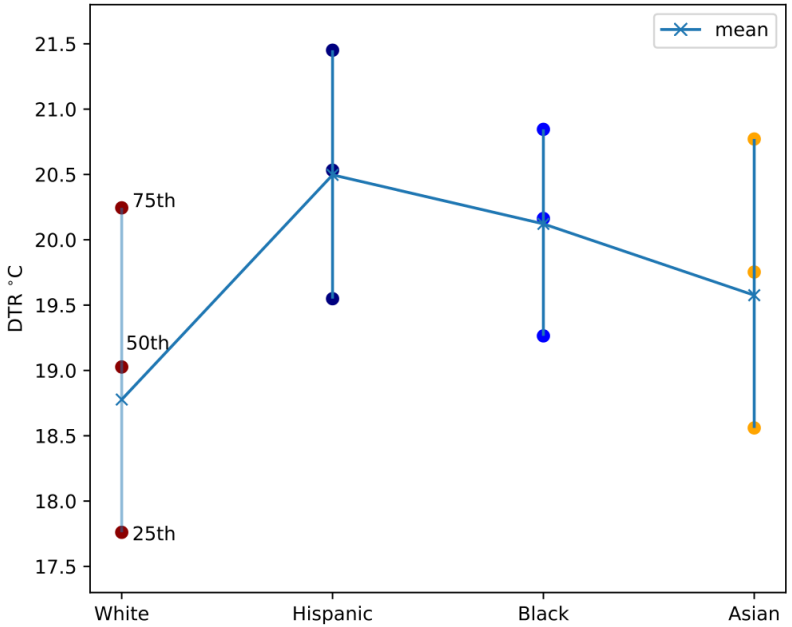

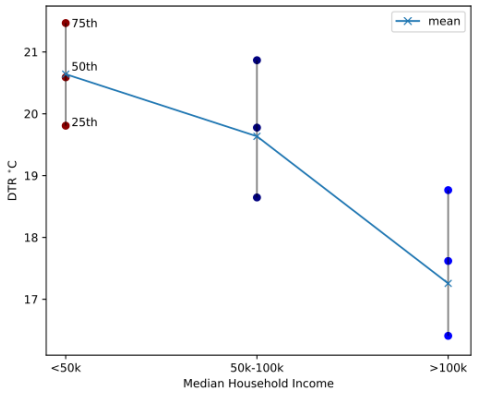

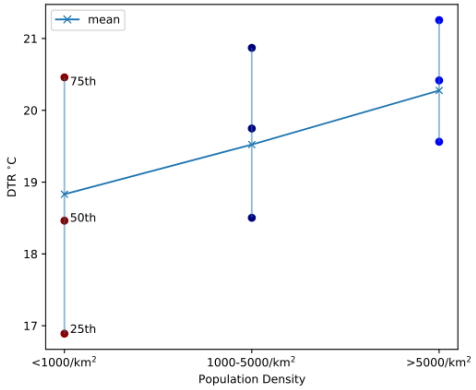

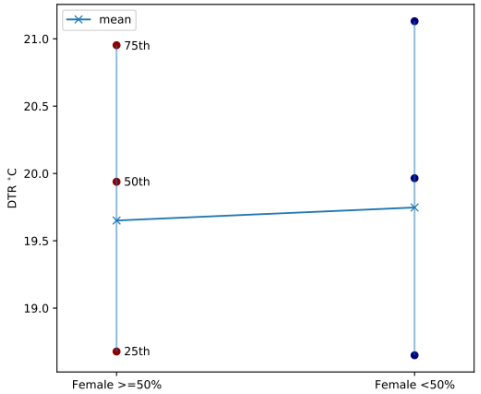

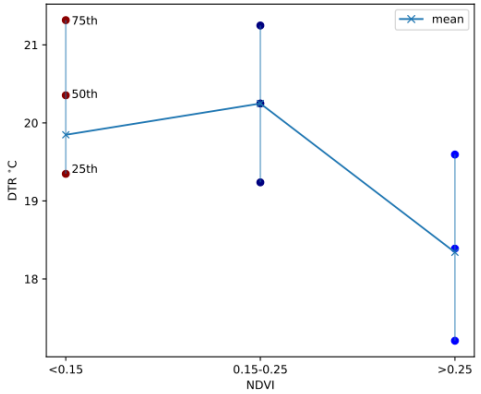

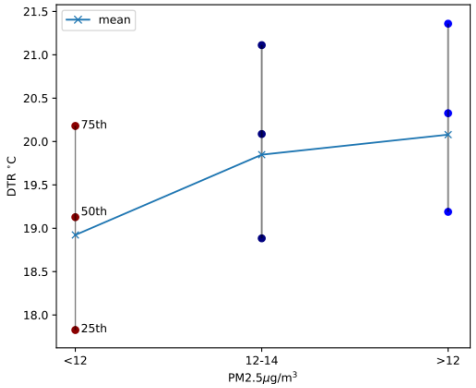

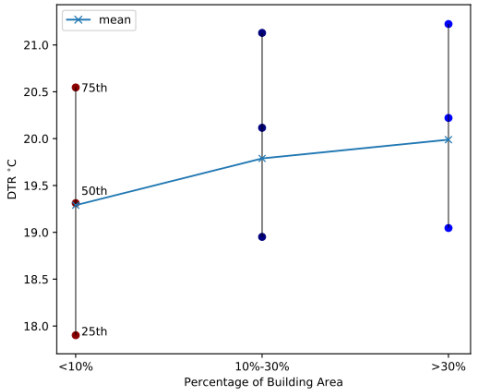

Satellite data for local insights and action

Unequal DTR exposure and mortality risk within the same city (Los Angeles)

Unequal exposure by race/ethnicity and income

Satellite data for local insights and action

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Is this unequal exposure across the U.S.?

Satellite data for local insights and action

Unequal DTR exposure and mortality risk within the same city (Los Angeles)

Unequal exposure by race/ethnicity and income

Is this unequal exposure across the U.S.? Yes

Unequal DTR exposure in the U.S.

[Liu and Smith-Greenaway 24]

Temperature:

- MODIS nighttime & daytime 2000-2017 monthly

Demographic and socioeconomic data:

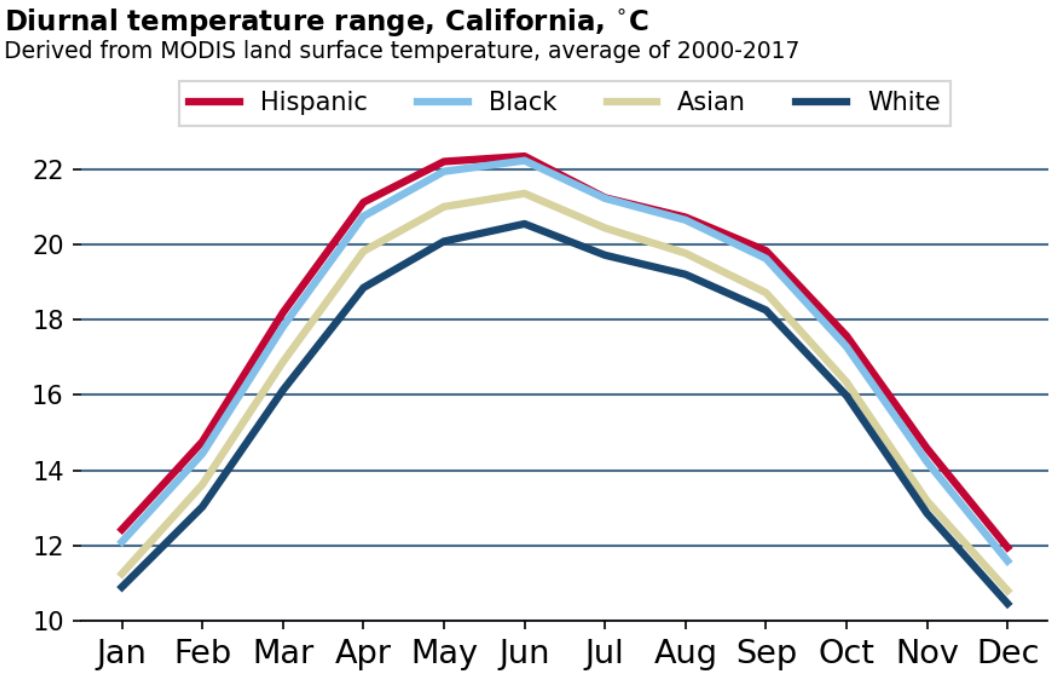
- American Community Survey (ACS) 2017
- census tract
- race/ethnicity, income, age, rental

Method:

- population-weighted exposure
- two-sample KS test

Unequal DTR exposure in the U.S.

[Liu and Smith-Greenaway 24]



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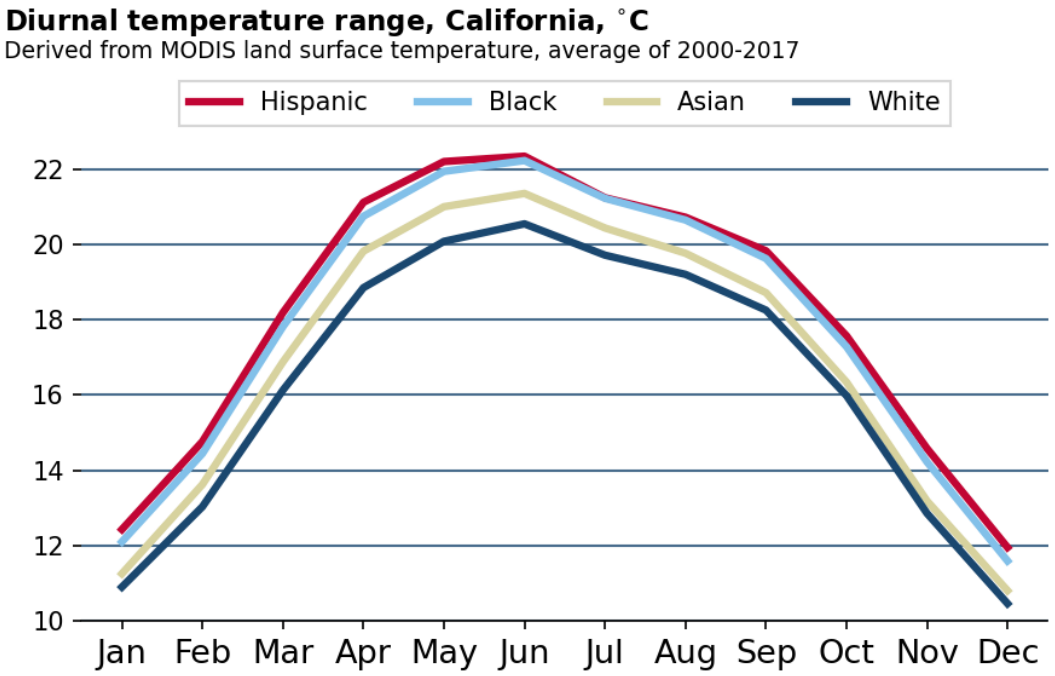
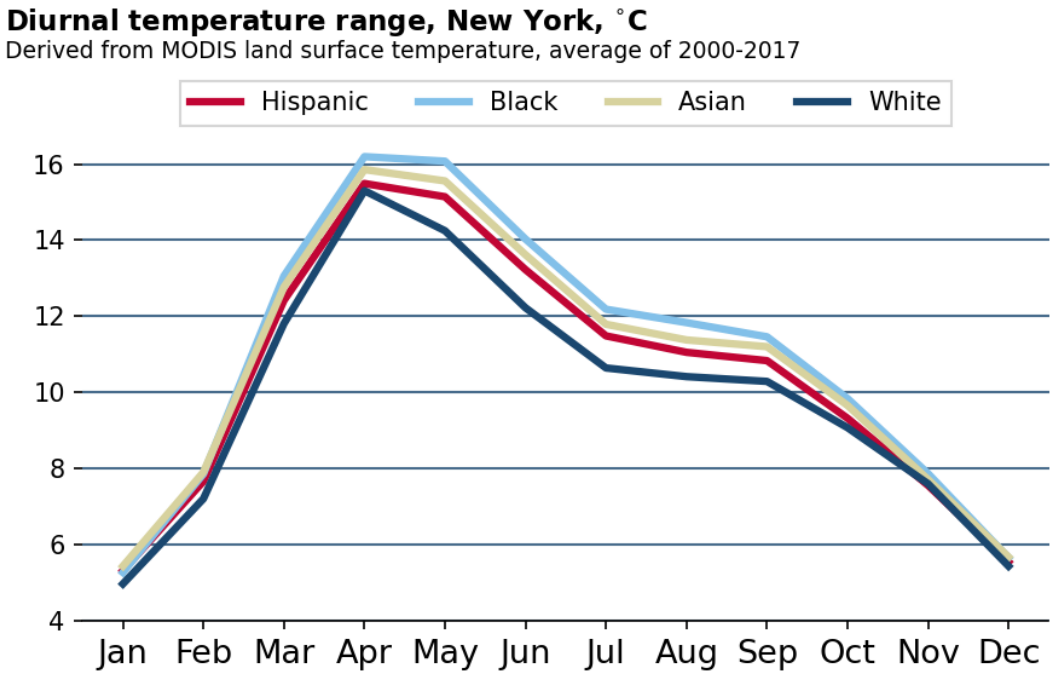
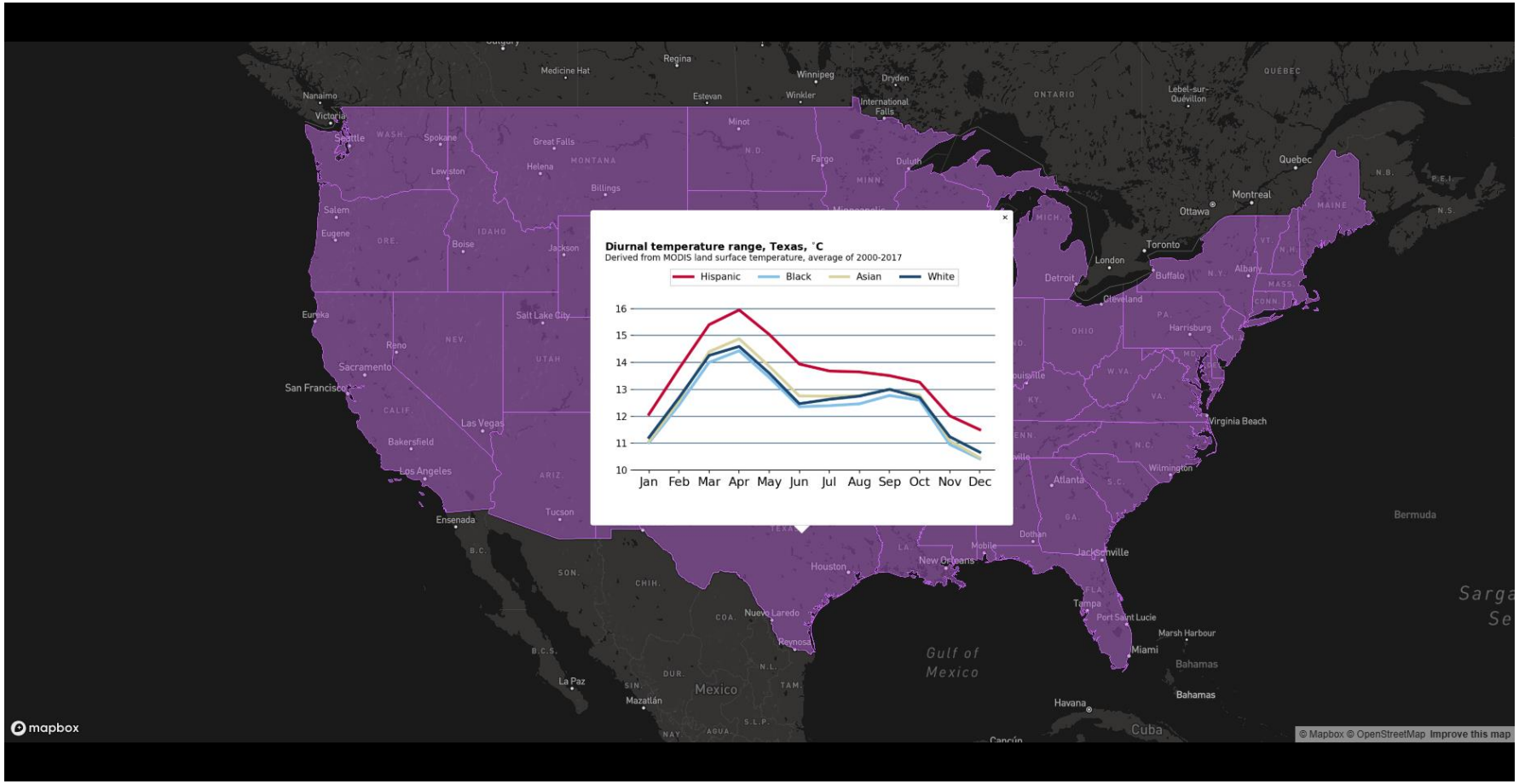




Unequal DTR exposure in the U.S.

live website: https://skrisliu.com/dtvus/map.html



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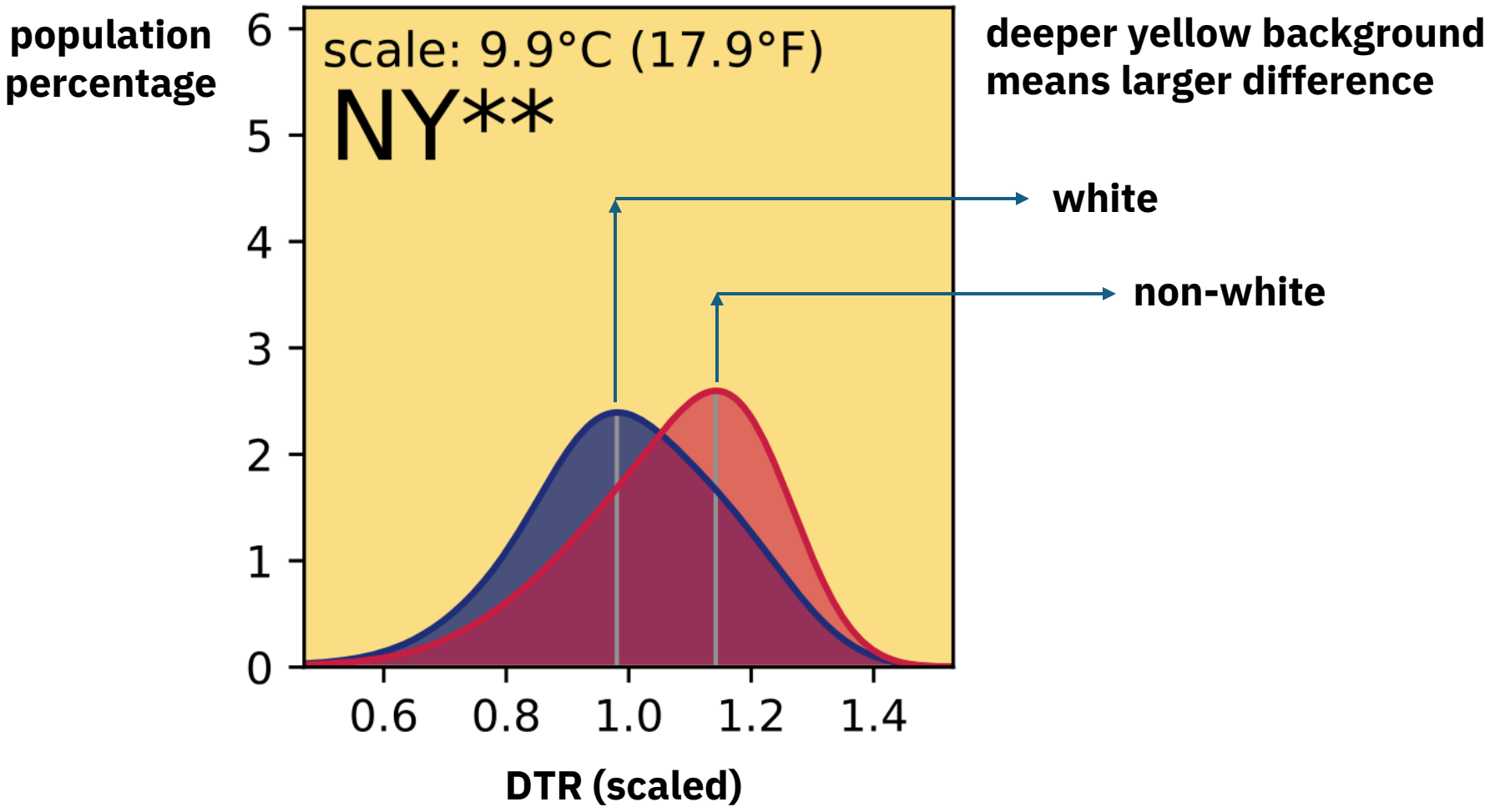
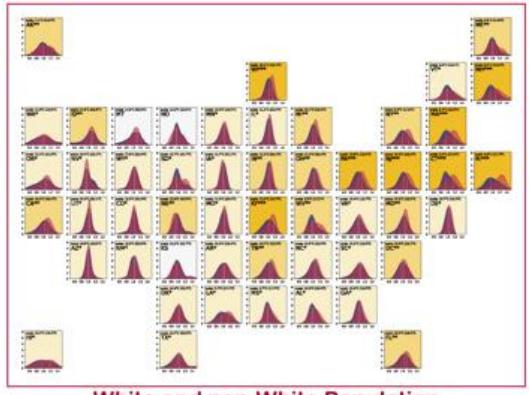
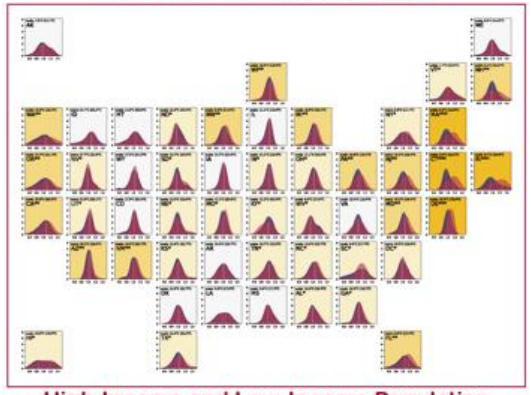
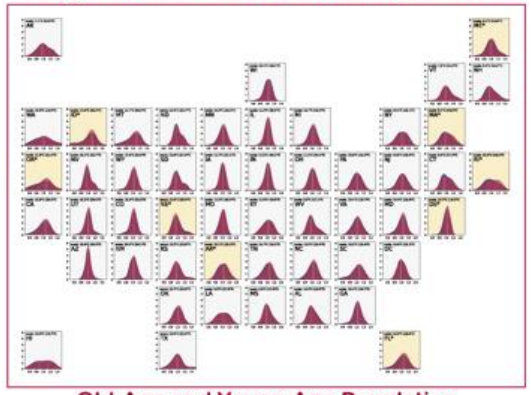




White and non-White Population



High-Income and Low-Income Population



Old-Age and Young-Age Population

race/ethnicity:	47 states + DC
income:	39 states + DC
age:	9 states

Satellite data is powerful, but .....

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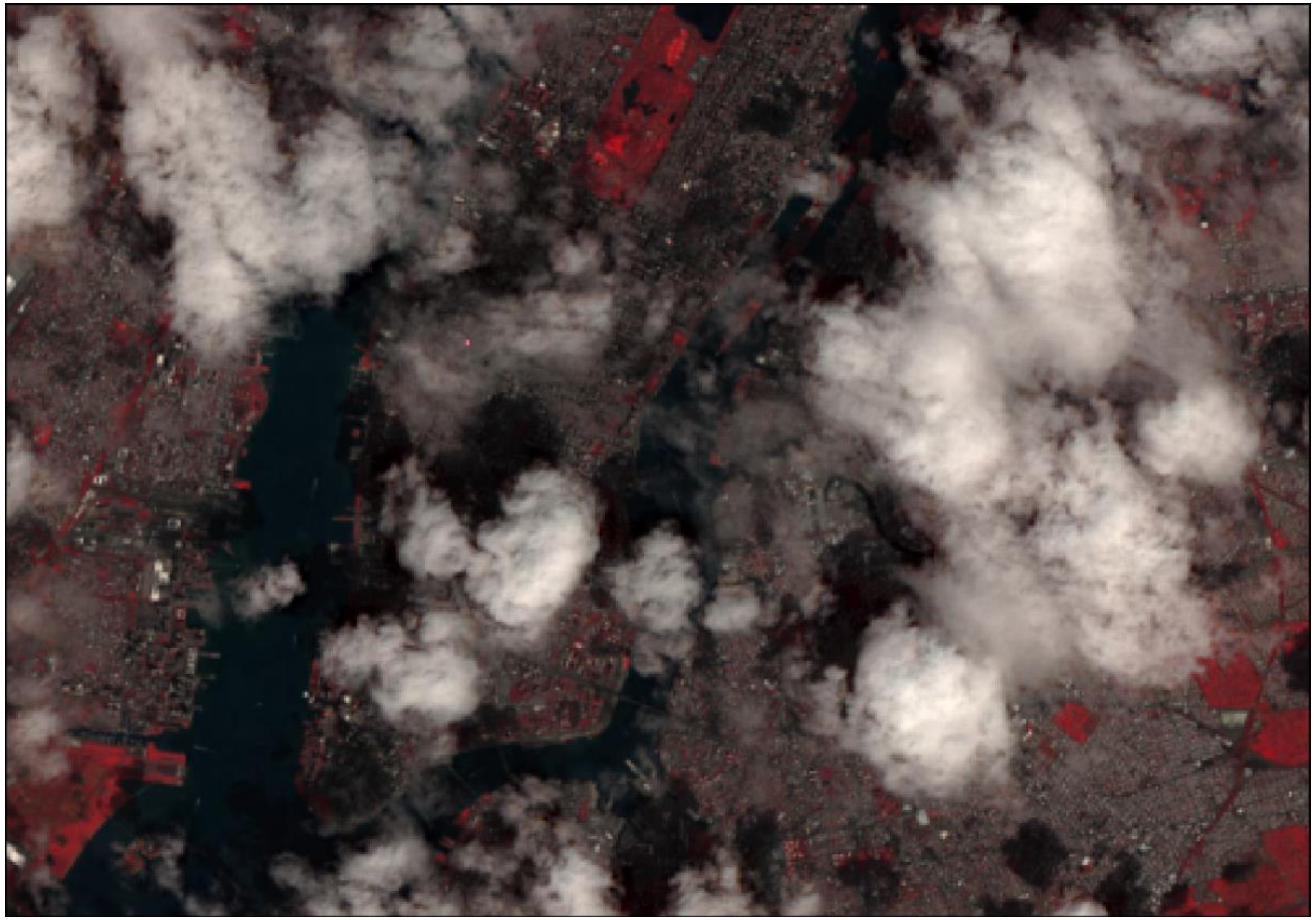
[Liu and Ho 24]



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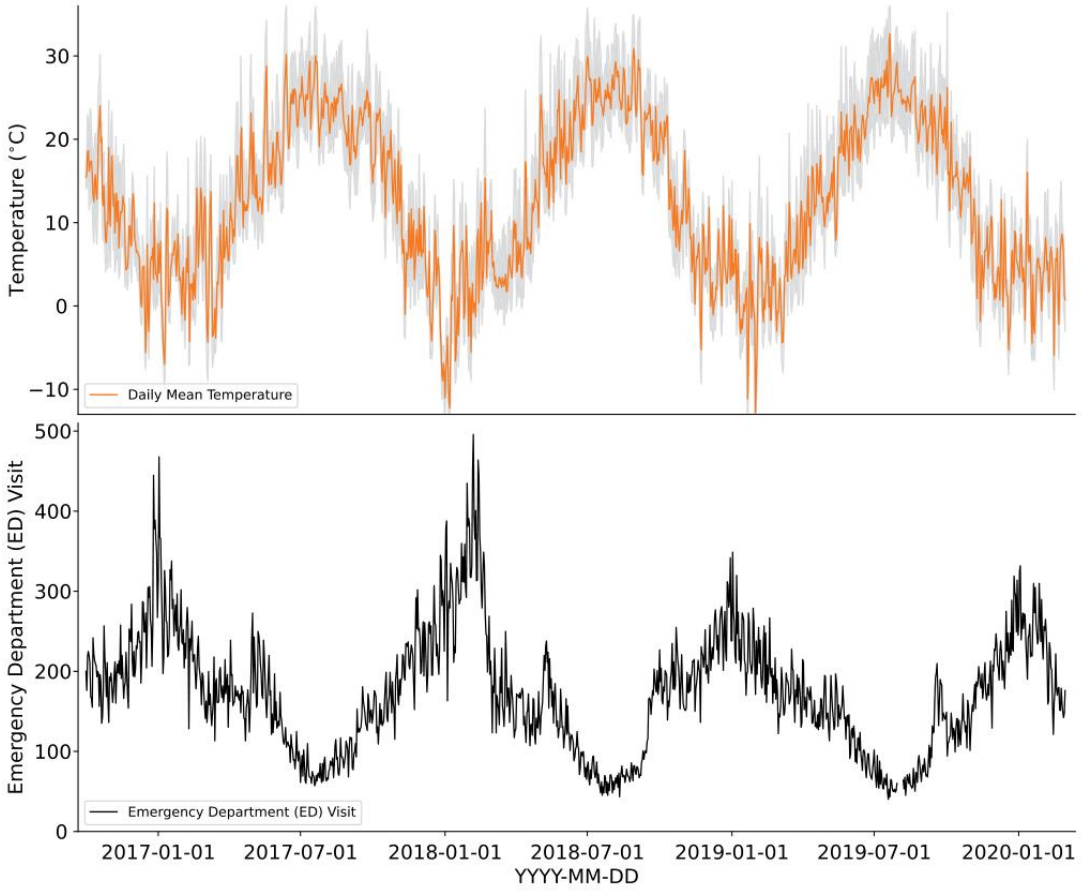
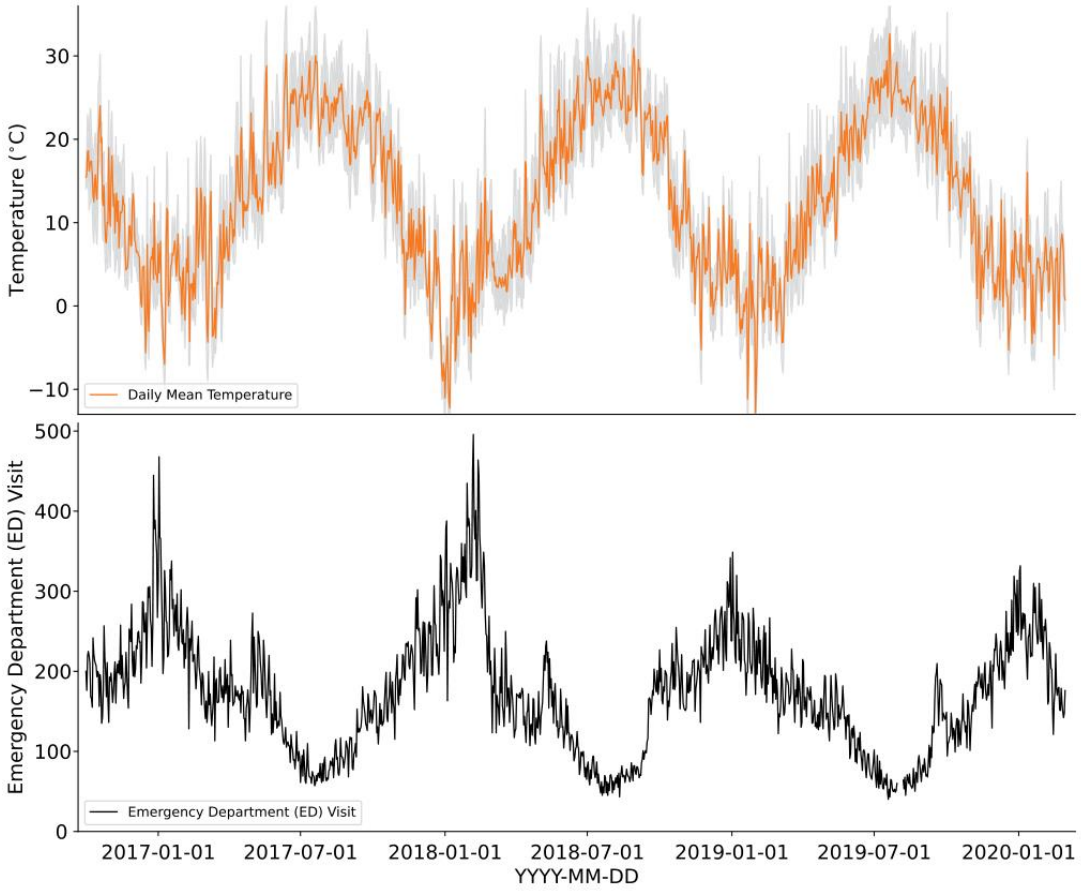


NYC Health releases daily counts of emergency department visit (zip code level)

Need daily temperature data for timeseries analysis

Satellite data is powerful, but .....

[Liu and Ho 24]



NYC Health releases daily counts of emergency department visit (zip code level)

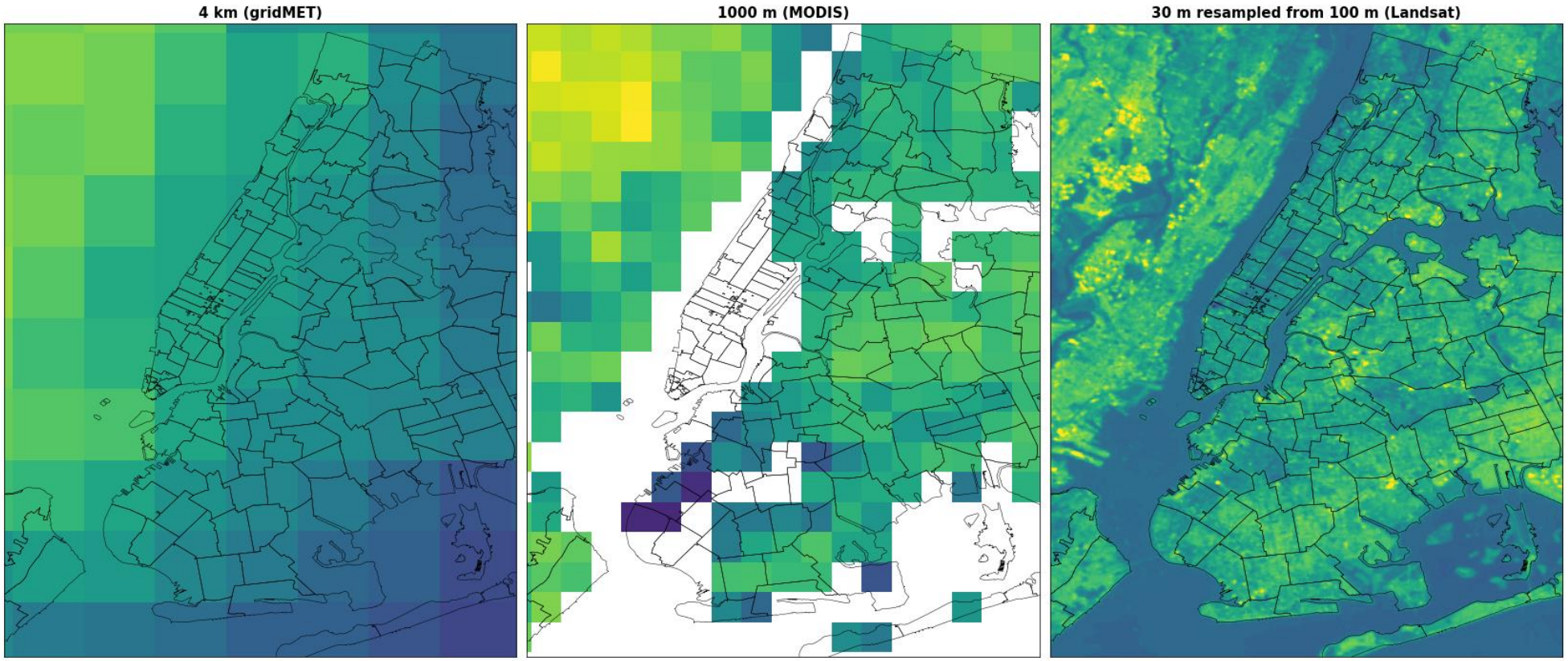
Need daily temperature data for timeseries analysis

Temperature data requirements

- daily (temporal domain)
- zip code level (spatial domain)

Satellite data is powerful, but .....

[Liu and Ho 24]



current best available data used in environmental health studies [Garcia et al. 23]

one image every 16 days (clouds, 66%)

Satellite data is powerful, but .....

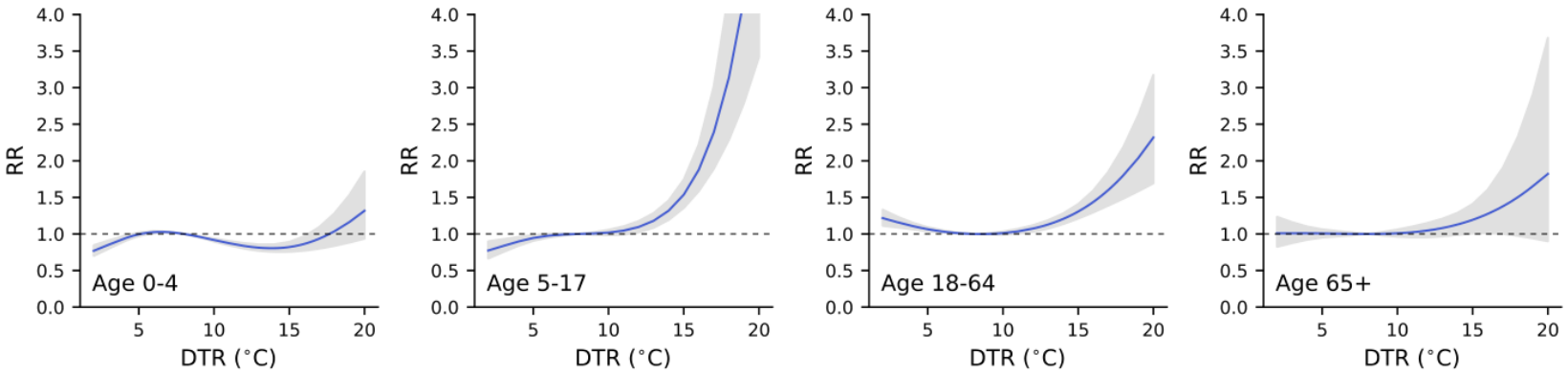
[Liu and Ho 24]

No satellite data available for

zip code level, daily temperature

Three meteorological stations from airports.
Did not use satellite data

[Liu and Ho 24]



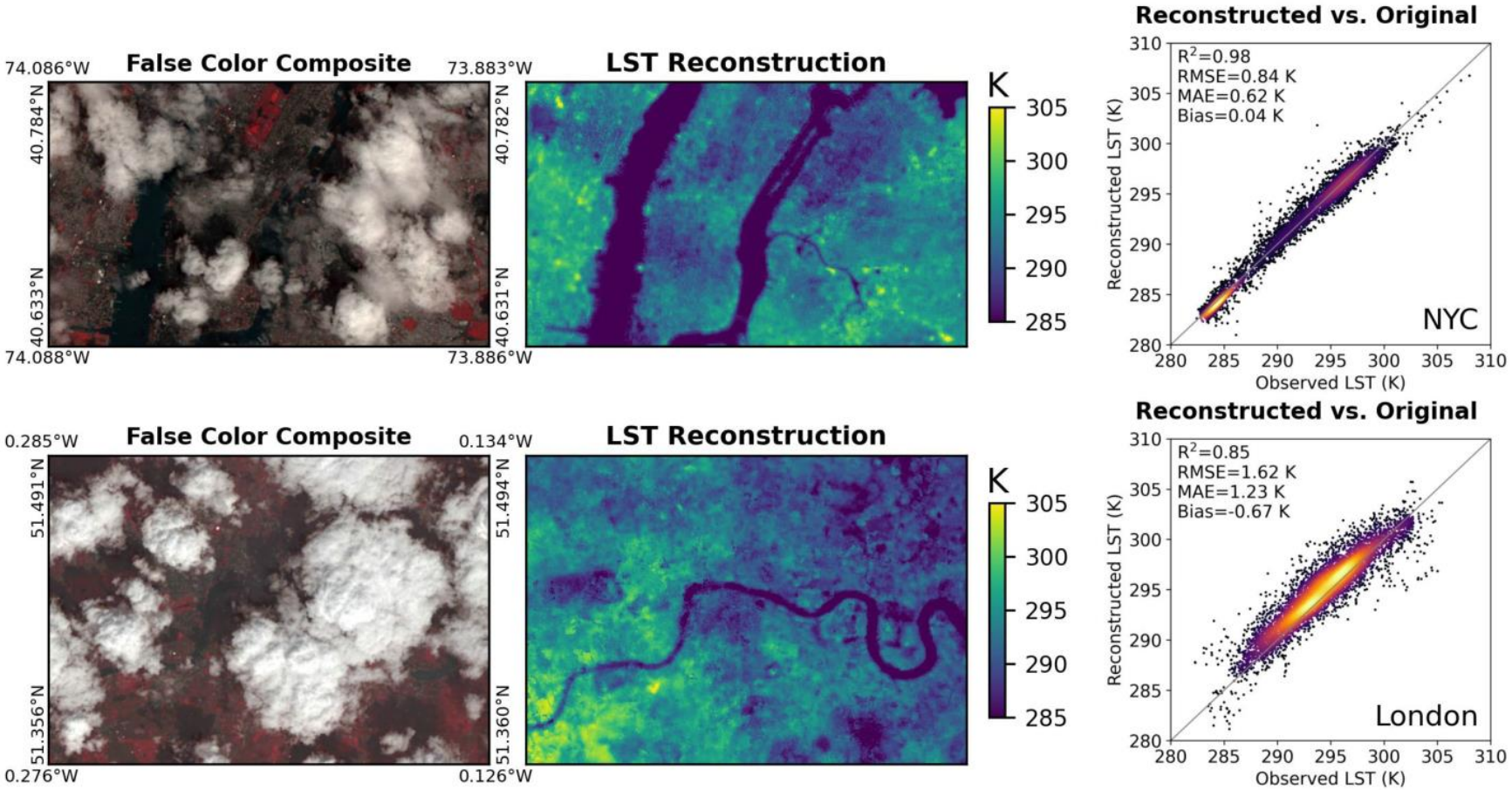
modeled via distributed lag nonlinear model (DLNM)

Satellite data & computing

Generate better temperature data

Beyond climate and health

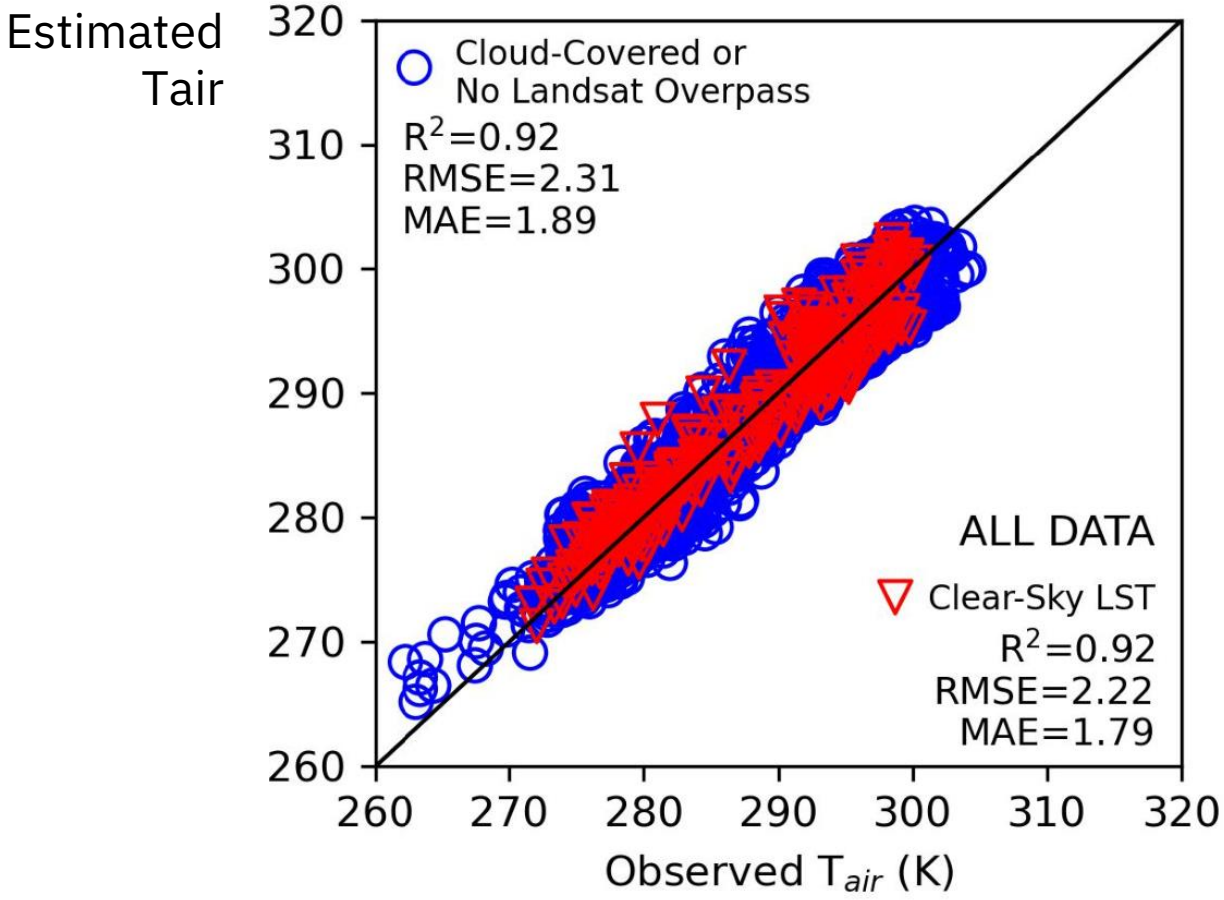
Daily temperature data finer than zip-code



[Liu et al. Under review]

Daily temperature data finer than zip-code

Validation with air temperature in NYC



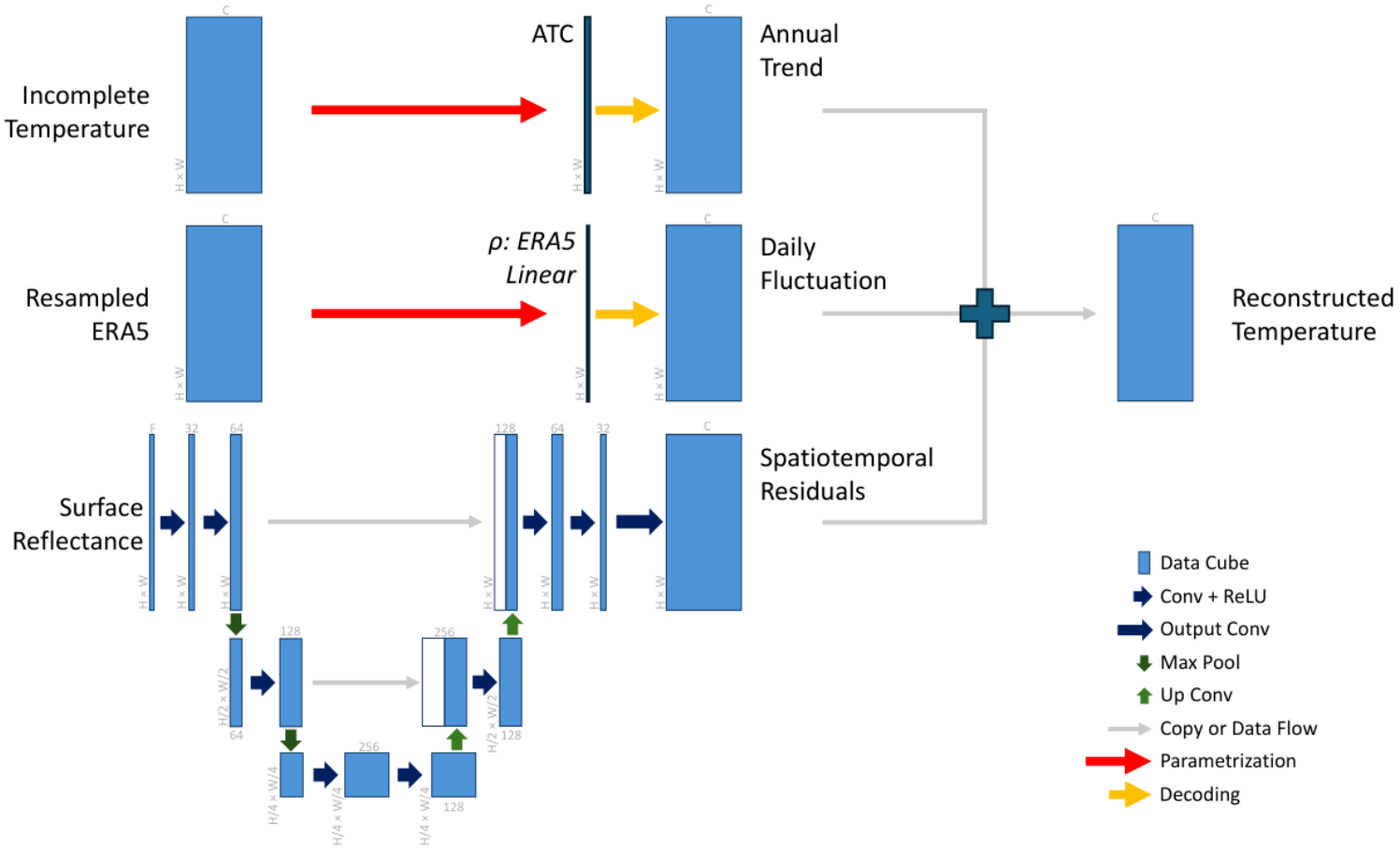
no difference between reconstruction and valid-observed data

after simple linear regression (adjusted for NDVI, elevation, solar radiation, solar zenith angle)

[Liu et al. Under review]

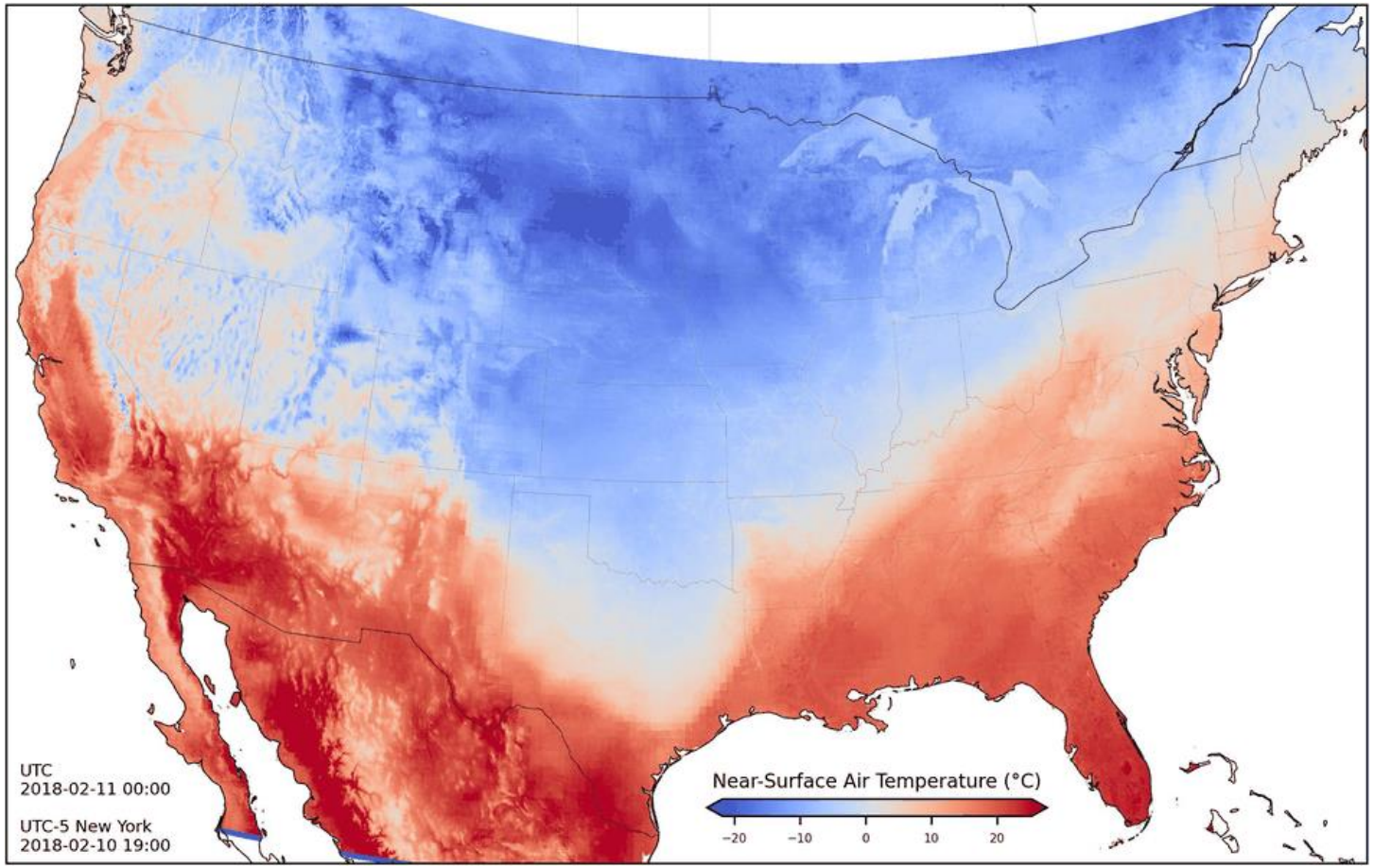
Daily temperature data finer than zip-code

self-supervised, physics-guided deep learning



[Liu et al. ICML 2025 workshop, Liu et al. ICCV 2025 workshop proceedings]

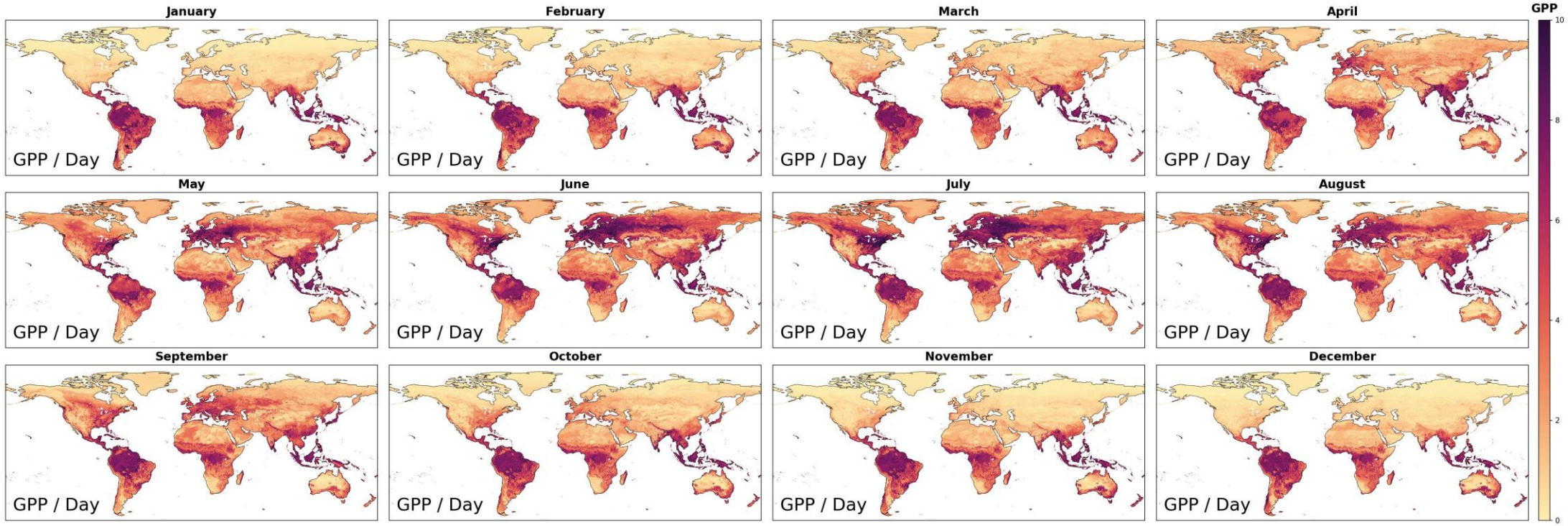
Hourly surface air temperature (2 km) USA 2018-2024



[Liu et al. Under review]

Global GPP & NEE mapping

with not satellite observations
input: only weather (temp, precipitation, solar), land (vegetation type), fluxnet



[Liu et al. Working paper]

Global GPP & NEE mapping

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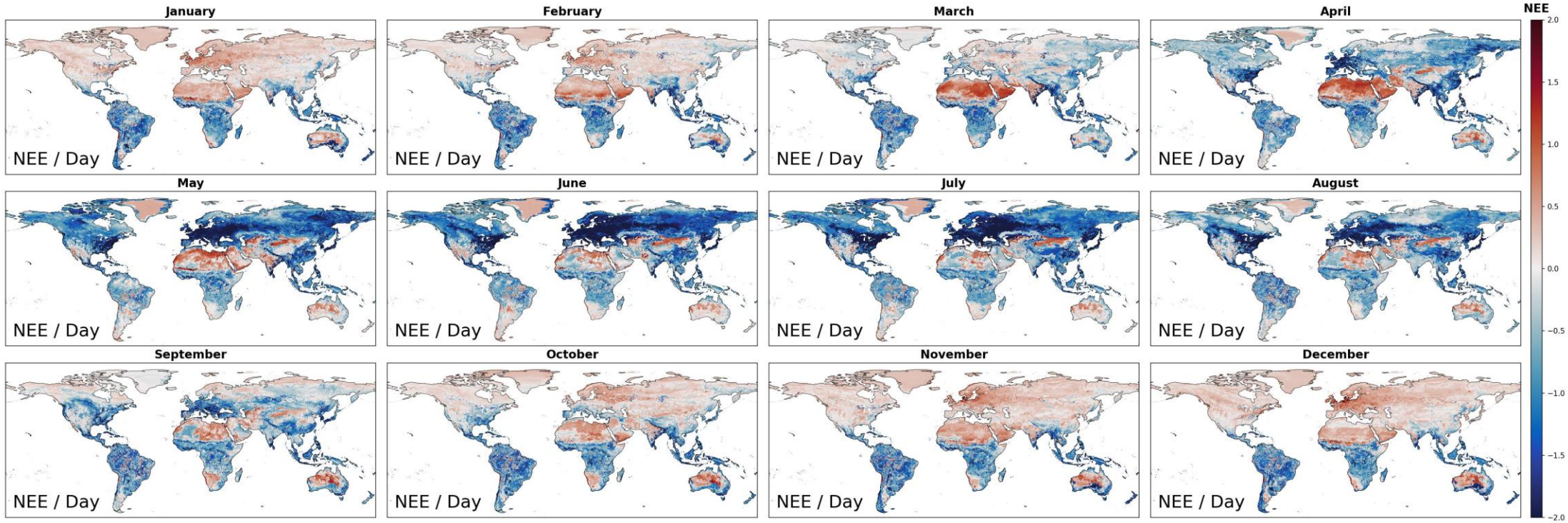
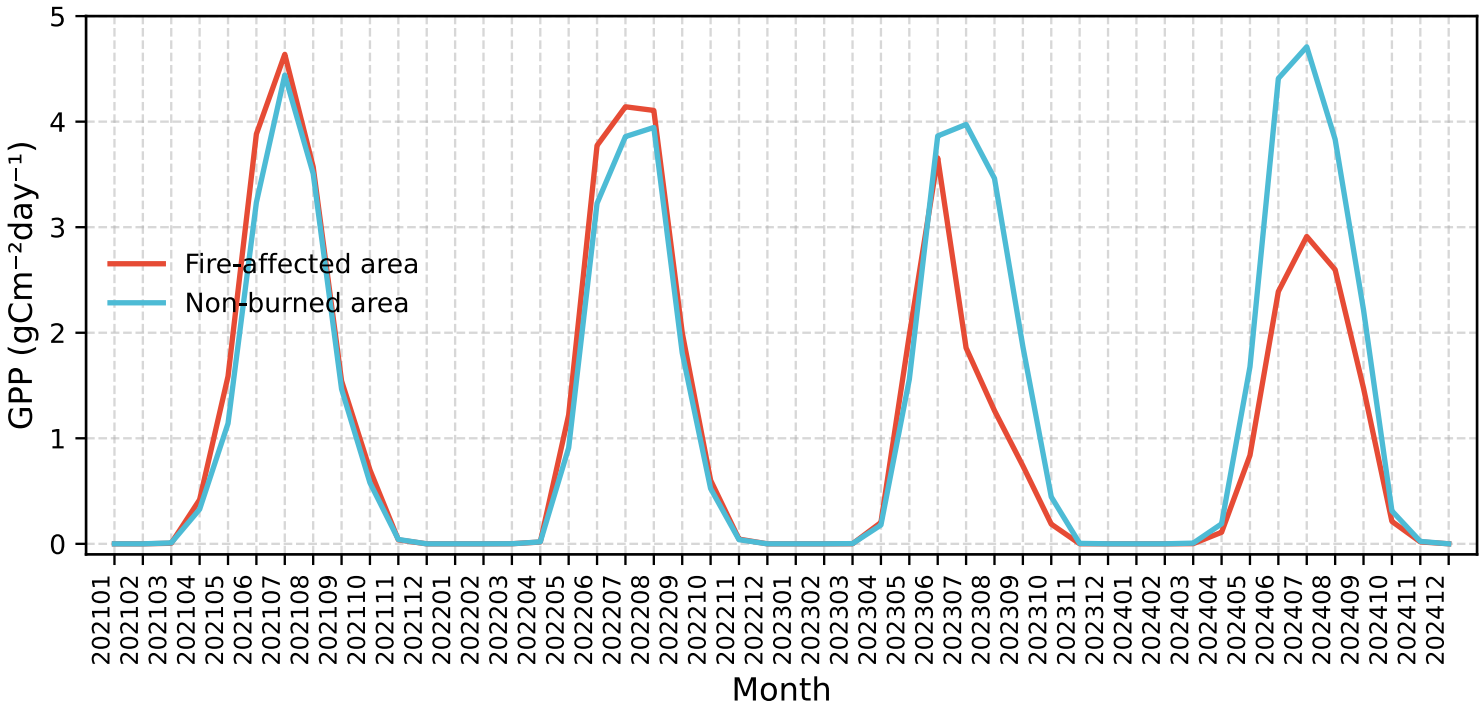


[Liu et al. Working paper]

Opportunity cost of wildfires
carbon intake reduction

Immediate carbon emissions: **647 TgC (570–727 TgC)** [Byrne et al., 2024]
Opportunity cost: **98 TgC**, or **15%** of immediate carbon emissions



[Liu et al. Working paper]

Satellite data is powerful

Real-time event detection

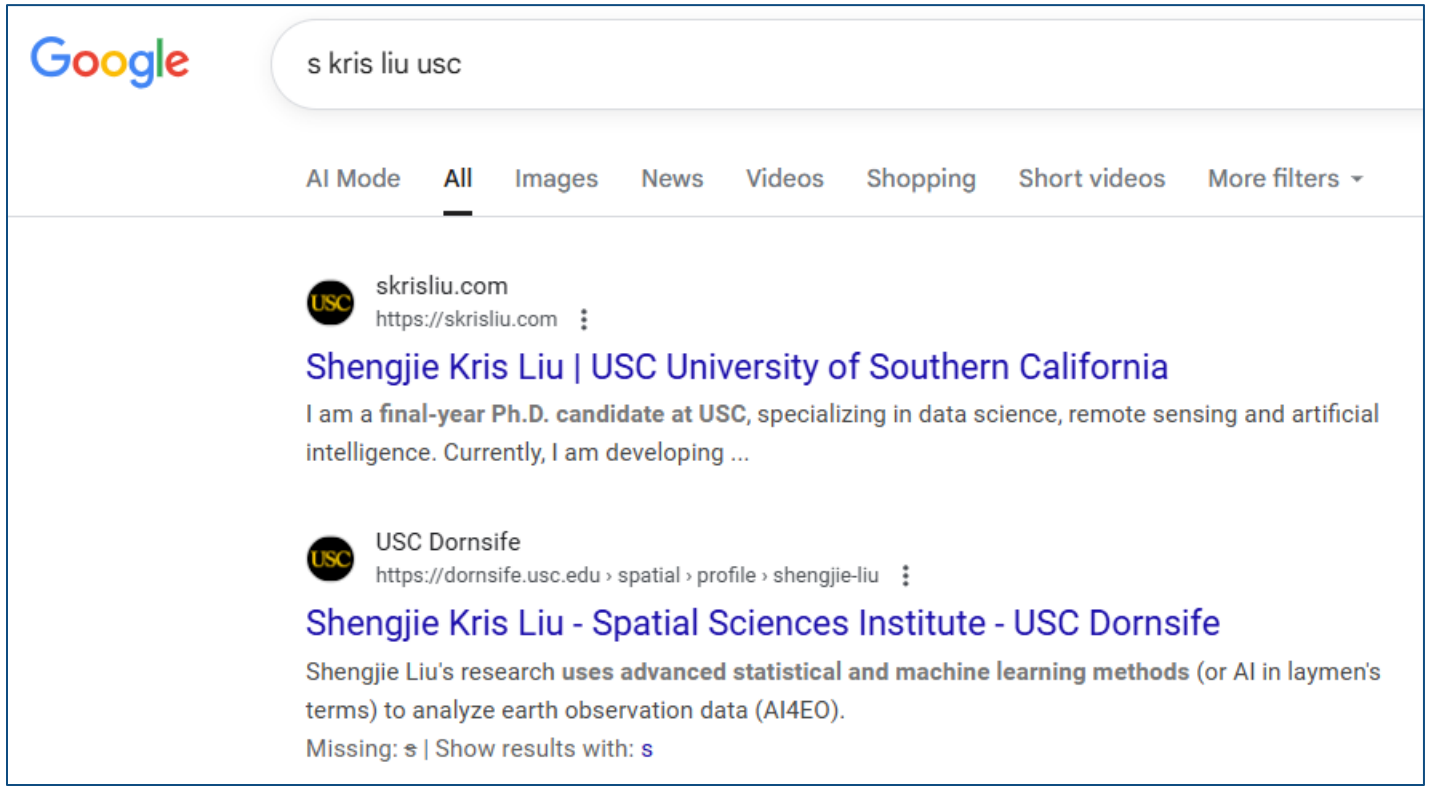
Provides global coverage, from high- to low-income regions

Cost-effective after initial setup

Growing public and private investment; becoming part of everyday life

Satellite data is not perfect

Complex processing	→ **Remote sensing scientist**
Calibration & accuracy	→ Combining with *in situ* data
Clouds	→ Algorithm (computing)
Sometimes long revisit time	→ Growing potential + algorithm

Thank you for your attention!

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